



CATHOLIC UNIVERSITY OF EASTERN AFRICA
PROJECT PROPOSAL FOR FINAL YEAR STUDY IN
Computer Science

BY

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1049507

PROJECT TITLE

STUDENTS' GRADUATION CLEARANCE SYSTEM
FOR TANGAZA UNIVERSITY

DATE: [MARCH, 2026]

Submitted in partial fulfillment of the requirement for the Bachelor of Science Degree in
Computer Science

Declaration and Approval

Student

I, Abigael K. Kirimi, declare that this proposal is my original work and that it has not been presented in any other university or institution for academic credit.

Signature:

Date:

Supervisor's Approval

This project proposal has been submitted for examination with my approval as university supervisor.

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Date:

Name:

Acknowledgment

First and foremost, gratitude is extended to God Almighty for the strength, wisdom, and grace that have sustained this academic journey.

Sincere appreciation is owed to the project supervisor for the patient guidance, constructive feedback, and consistent encouragement that shaped this proposal from its earliest stages. The support provided has been invaluable.

Gratitude is also expressed to family members whose sacrifices and steadfast support made the pursuit of this degree possible. This work is dedicated to them.

Finally, the administration and staff of Tangaza University -in particular the Library, Finance Office, Student Affairs Office, and the Academic Registrar's Office -are warmly acknowledged for generously sharing their time and insight into the current clearance process. Their contributions were essential to identifying the problem this system seeks to solve.

Abstract

Graduation clearance remains one of the most stressful and time-consuming administrative processes for final-year students in many Kenyan universities. At Tangaza University, a Catholic institution based in Nairobi, this process is entirely manual. Students must physically visit six separate offices -the Director, the Library, the Finance Office, the Dean's Office, the University store and the Office of the Academic Registrar -to have a paper form signed and approved before they can graduate. This results in long queues, misplaced forms, repeated trips, and considerable frustration, particularly for students who are simultaneously sitting final examinations, completing research projects, or managing employment obligations.

This proposal presents the design and development of a Students' Graduation Clearance System (SGCS) for Tangaza University. The SGCS is a web-based application that will automate and centralise the entire clearance process. Students will be able to log in, initiate their clearance online, monitor the approval status of each department in real time, and download a digital clearance certificate once all required approvals have been obtained. The system incorporates a structured multi-departmental workflow in which the student's clearance request is processed sequentially through the Director, Library, Finance Office, Dean, and Academic Registrar. Departmental officers have a dedicated login portal through which they can review, approve, or reject clearance requests and communicate feedback to students. A transcript request and upload module, financial clearance verification, and departmental reporting functionality are also included.

The system is built using standard web technologies: HTML, CSS, and JavaScript on the client side, and PHP for server-side scripting, supported by a relational database for centralised data management. The development follows the Structured System Analysis and Design (SSAD) methodology, with requirements gathered through structured interviews and questionnaires. The outcome of this project is expected to significantly reduce clearance processing time at Tangaza University and improve the graduation experience for all stakeholders.

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List of Terms and Abbreviations

CSS	Cascading Style Sheets. A style sheet language used to describe the presentation of HTML documents.
DFD	Data Flow Diagram. A graphical representation of how data flows through a system.
ERD	Entity Relationship Diagram. A modelling tool used in database design.
Director	Director.
HTML	HyperText Markup Language. The standard language for creating web pages.
ICT	Information and Communication Technology.
JKUAT	Jomo Kenyatta University of Agriculture and Technology.
JS	JavaScript. A scripting language for dynamic web content.
NF	Normal Form. A stage in database normalisation.
PHP	Hypertext Preprocessor. A widely used server-side scripting language for web development.
SGCS	Students' Graduation Clearance System.
SMTP	Simple Mail Transfer Protocol. The protocol used for sending email.
SSAD	Structured System Analysis and Design. A system development methodology.
UAT	User Acceptance Testing. The final phase of system testing involving real users.
UI/UX	User Interface / User Experience.
UoN	University of Nairobi.

CHAPTER ONE: INTRODUCTION

1.1 Background of the Study

The emergence and rapid expansion of Information and Communication Technology (ICT) has fundamentally reshaped the way institutions operate across every sector of society. From healthcare and banking to government administration and retail, digital systems have made it possible to process information faster, more accurately, and with considerably less reliance on paper-based procedures than was conceivable even two decades ago. The benefits of this shift are well documented: reduced processing times, improved accuracy, enhanced transparency, and greater accessibility for the people these institutions serve.

In higher education, this transformation has been particularly pronounced. Universities around the world have progressively replaced manual administrative systems with computerised alternatives for functions such as student registration, fee collection, examination scheduling, and results processing. In Kenya, this transition gained significant momentum in the early 2010s, driven by national ICT policy initiatives and the rapid spread of affordable internet access. Institutions such as the University of Nairobi, Kenyatta University, and JKUAT led the adoption of online student portals that allow learners to access administrative services remotely. Research by Mwangi and Otieno (2021) found that Kenyan universities which introduced digital administrative systems recorded processing time reductions of up to sixty percent, alongside measurable improvements in record-keeping accuracy. Munyua (2020) further established that administrative staff in automated institutions reported significantly reduced time spent on repetitive manual tasks, freeing capacity for higher-value work.

Graduation clearance is the formal process through which a final-year student confirms with each relevant department that all outstanding academic, financial, and library obligations have been resolved before a degree certificate and academic transcript can be released. The typical clearance workflow requires the student to obtain approval from the Director, the Library, the Finance Office, the Dean, and finally the Academic Registrar, each of whom must verify that the student meets the requirements for graduation from their respective standpoint. In many Kenyan universities, this process still relies on the physical circulation of a paper form from one office to the next, with each officer signing in turn. The limitations of this approach are significant: officers are not always available when students arrive, queues can be long, forms are sometimes misplaced, and the process can extend across several days or even weeks.

Tangaza University, a Catholic institution of higher learning located along Langata Road in Nairobi, Kenya, is among the institutions where graduation clearance remains entirely manual. There is currently no digital system to support or track this process. Students must make multiple physical visits to six separate offices, and there is no mechanism for checking clearance status without appearing in person at each department. This creates an experience that is unnecessarily stressful, particularly for students who are simultaneously sitting final examinations, completing research projects, or managing employment commitments alongside their studies.

This study proposes the development of a Students' Graduation Clearance System (SGCS) for Tangaza University. The SGCS is a web-based application designed to automate and centralise the entire clearance process. Students will be able to log in, initiate clearance online, and monitor progress across all six departments in real time, receiving notifications at each stage. Departmental officers will access a dedicated interface through which they can review, approve, or reject clearance requests. The Academic Registrar will have access to comprehensive reports on clearance progress institution-wide, and a transcript request and upload module will allow relevant documentation to be handled within the same system. The proposed system is built using standard web technologies, HTML, CSS, and JavaScript on the client side; PHP for server-side scripting; and MySQL for centralised data management, ensuring broad compatibility and maintainability.

1.2 Problem Statement

The graduation clearance process at Tangaza University is currently carried out entirely on paper and requires final-year students to physically visit six separate departments before their graduation can be confirmed. In practice, this process frequently extends across several days due to a combination of compounding difficulties. Officers are not always present when students arrive, requiring multiple return visits to the same office. There is no system to track which departments have been cleared and which are still pending, meaning students must travel to campus in order to discover information that a digital system could provide instantly. When a student encounters a problem, such as an outstanding library fine or an unresolved fee balance, they typically learn of it only after wasting time and transport costs making a trip to the relevant office.

From the perspective of institutional administration, the paper-based system creates equally serious problems. The Academic Registrar's office has no reliable way to determine, at any

given point in time, how many students have completed clearance and how many remain in progress. There is no centralised audit trail, which means that resolving disputes, for example, where a student maintains they obtained clearance from a particular department but no signed form can be produced, is both time-consuming and difficult to adjudicate fairly. Nakamura and Lohr (2019) observed that institutions without integrated clearance systems impose unnecessary administrative burdens on their staff, who must devote significant time to coordination tasks that an automated system could handle without human intervention.

Data integrity is a further concern with the current system. Paper forms can be lost, damaged, or duplicated. Information recorded on one form may be inconsistent with records held within a department, and there is no mechanism for cross-referencing data across departments in real time. This creates conditions under which errors can go undetected and where the accuracy of institutional records cannot be reliably guaranteed. Communication between the university and its graduating students about the status of their clearance is similarly limited: without a digital system, reminders, rejection notices, and completion confirmations must be issued manually, and in many cases are not issued at all. The cumulative effect of these deficiencies, inefficiency, lack of transparency, poor communication, data integrity risks, and the absence of a centralised audit trail, constitutes a significant institutional problem that the proposed Students' Graduation Clearance System is designed to address.

1.3 Aim of the Research

To design and develop a web-based Students' Graduation Clearance System that automates and streamlines the multi-departmental graduation clearance process at Tangaza University.

1.4 Specific Objectives

The following specific objectives will guide the achievement of the main objective:

1. To analyse the current manual graduation clearance process at Tangaza University through structured interviews and questionnaires, identifying its limitations and the requirements for the proposed system.
2. To design the proposed Students' Graduation Clearance System, including the database structure, process flows, and user interface prototypes, guided by the SSAD methodology.

3. To develop the core system functionalities, including the multi-departmental sequential clearance workflow, the transcript request and upload module, financial clearance verification, and departmental reporting for the Director and the Academic Registrar.
4. To implement real-time status tracking and automated notification features that keep students and officers informed at each stage of the clearance process.
5. To test the developed system through developer testing, end-user testing, and heuristic evaluation, and to deploy the system within the Tangaza University environment.

1.5 Justification

The development of the Students' Graduation Clearance System is justified on several grounds, each anchored in existing research and the demonstrable operational needs of Tangaza University.

The most immediate justification is the evidence that digital clearance systems deliver measurable efficiency gains. Mwangi and Otieno (2021) found that Kenyan universities that adopted automated administrative systems achieved reductions in processing time of up to sixty percent. Implementing a comparable system at Tangaza University would allow clearance to be completed in a fraction of the time currently required, reducing the burden on both students and staff. The SGCS eliminates the need for physical visits to multiple offices, making the process accessible from any internet-connected device, including a smartphone, at any time. This is a significant benefit for students who live far from campus or who are managing work and study simultaneously.

Kariuki (2020) established that the stress associated with manual clearance processes has a measurable negative impact on student wellbeing during an already demanding period of their academic career. A system that provides real-time visibility into clearance progress and sends automated notifications at each approval stage addresses this concern directly by removing the uncertainty and repeated trips that generate much of the stress the current process involves.

From an institutional perspective, the SGCS enhances transparency and accountability by creating a centralised, auditable digital record of every clearance request, approval, and rejection. This eliminates the risk of lost or duplicated forms and enables the Academic Registrar to generate accurate, up-to-date reports on clearance progress institution-wide. The system also supports Kenya's broader national agenda of digital transformation in public

services, contributing to the objectives articulated in Kenya Vision 2030 for technology-driven improvement in service delivery.

Finally, the decision to build the system on widely used, well-documented standard web technologies; HTML, CSS, JavaScript, PHP, and a relational database; ensures that the system is maintainable and extensible by staff with standard web development skills, without dependence on proprietary platforms or expensive commercial software licences.

1.6 Scope

This study focuses specifically on the graduation clearance process at Tangaza University, Nairobi. The SGCS will cover six departments involved in the clearance workflow: the Director, the Library, the Finance Office, the Dean's Office, the University store and the Office of the Academic Registrar. The system will not extend to other academic processes such as examination management, student registration, or fee payment processing, as these fall outside the boundaries of the graduation clearance workflow.

The target users of the system are final-year undergraduate students and the administrative officers in each of the six clearance departments. The features to be implemented include: student login and online clearance initiation; a sequential multi-departmental approval workflow; a transcript request and upload module; financial clearance verification; real-time clearance status tracking with automated notifications; departmental reporting accessible to the Director for department-level data and to the Academic Registrar for institution-wide clearance reports; and digital clearance certificate generation for students who have obtained all required approvals. The system will be developed using HTML, CSS, JavaScript, PHP, and MySQL. The project timeline runs from February to August 2026, as detailed in the project schedule presented in Chapter Four.

1.7 Research Organisation

This research is organized into chapters. Chapter Two presents a review of five related systems from global and local contexts, a historical overview of student clearance systems in higher education, emerging trends in the field, and the research gap this project addresses. Chapter Three describes the methodology that will guide the development of the system, covering requirement elicitation, system analysis, design, implementation, testing, and deployment, with all relevant analytical diagrams and design artefacts included.

CHAPTER TWO: LITERATURE REVIEW

2.1 Introduction

This chapter reviews existing literature and related systems that are relevant to the proposed Students' Graduation Clearance System for Tangaza University. It begins with a historical overview of how student clearance and administrative systems have evolved in higher education. Five related systems are then reviewed in detail, with the strengths and weaknesses of each discussed and compared against the proposed SGCS using comparative tables. The chapter subsequently examines emerging trends in university information systems and concludes with a summary of the findings from the reviewed systems. This review provides the evidence base that informs the design decisions presented in Chapter Three.

2.2 History of the Research Topic

The concept of automating university administrative processes has a history extending back to at least the 1980s, when universities in North America and Western Europe began deploying computer-based student information systems to manage records such as course registration and grades. These early systems were largely text-based and accessible only to administrators; students had no direct interface with them. By the 1990s, the widespread adoption of the internet fundamentally altered this model, making it technically feasible for universities to build systems that students themselves could access directly, a development that transformed institutional expectations around service delivery.

In the 2000s, integrated university management platforms began to emerge, sometimes built on Enterprise Resource Planning (ERP) frameworks that brought together student registration, fee management, academic records, and library systems under a single digital environment. Universities in the United States and Europe adopted these platforms widely. In sub-Saharan Africa, however, adoption was considerably slower. Wanyembi (2018) identified cost, inadequate ICT infrastructure, and limited technical expertise as the primary barriers to adoption on the continent.

Kenya began making significant progress in the early 2010s, propelled by national ICT policy and improving internet accessibility. Larger public universities including the University of Nairobi, Kenyatta University, and JKUAT led the deployment of online student portals. Despite this progress, graduation clearance specifically remained an underserved area: even institutions with well-developed student portals frequently left clearance as a manual, paper-based exception. This gap is particularly pronounced at smaller private and faith-based

institutions like Tangaza University, where resources for large-scale ICT projects are more limited and the incentive to automate niche administrative processes is often deprioritised in favour of core academic systems.

2.3 Review of Related Prototypes and Systems

2.3.1 Case 1: University of Nairobi (UoN) Student Clearance Portal

The University of Nairobi integrated an online clearance portal into its existing Student Information System. The portal enables students to log in and submit clearance requests electronically. Each participating department receives a notification when a student's request arrives, and officers can approve or flag the request from within the system. A clearance certificate is generated for students who complete all departmental approvals.

The principal strength of this system is its integration with the university's central student database, which means that student details do not need to be re-entered at each stage, reducing the risk of data entry errors. The system also allows the Academic Registrar to monitor clearance progress across all departments from a single dashboard. However, Omondi (2020) identified two significant weaknesses. The portal is not mobile-responsive, which is a serious limitation given that most Kenyan students access online services via smartphone. Additionally, the system only notifies students by email upon final clearance, meaning students have no visibility into the status of individual departmental approvals as they happen. The proposed SGCS addresses both of these gaps through a responsive interface and per-department status notifications.

Table 1: Comparison of Proposed System with Case 1 (University of Nairobi)

S/N	Proposed System (SGCS)	Case 1: UoN Clearance Portal
1	Mobile-responsive design accessible via smartphone	Not mobile-responsive
2	Email notification at each departmental approval	Email notification only on final clearance
3	Real-time per-department status visible to student	Status not visible between submissions
4	Digital clearance certificate downloadable from portal	Certificate printed at the registrar's office
5	Sequential approval workflow with six department	Limited departmental workflow configuration

2.3.2 Case 2: Catholic University of Eastern Africa (CUEA) Student Portal

The Catholic University of Eastern Africa (CUEA), the parent institution of this project, operates a comprehensive student information portal that provides students with access to academic records, fee statements, course registration, and a range of other administrative services. Notably, the portal includes a dedicated Students' Graduation Clearance module through which graduating students can initiate and track their clearance process online. This makes CUEA one of the few faith-based institutions in Kenya to have implemented a digital clearance workflow within an existing student portal, and it provides a directly relevant point of comparison for the proposed SGCS.

The CUEA clearance module allows students to log in and view the clearance status of each participating department through a centralised dashboard. Departmental officers can access the system to update clearance records on their end, and the overall status of a student's clearance is visible within the portal. This represents a meaningful step beyond the fully manual, paper-based processes that characterise institutions such as Tangaza University, where no equivalent digital mechanism exists.

However, a review of the CUEA system reveals several limitations that the proposed SGCS is designed to address. The system does not send automated notifications to students at each stage of departmental approval, meaning students must log in and check their status manually rather than being informed proactively when an action is taken or when their attention is required. The system also does not include a transcript request and upload module integrated within the clearance workflow, meaning documentation-related steps must still be handled through separate channels. Additionally, the clearance certificate, once all approvals are obtained, is not generated digitally within the portal for immediate download but must be collected through a manual process. The proposed SGCS addresses each of these gaps by incorporating automated per-department notifications, an integrated transcript module, and digital certificate generation, all within a single unified system designed specifically for the Tangaza University context.

Table 2: Comparison of Proposed System with Case 2 (Catholic University of Eastern Africa)

S/N	Proposed System (SGCS)	Case 2: CUEA Student Portal
1	Automated notification to student at each departmental approval stage	No automated per-stage notifications; student must check status manually

S/N	Proposed System (SGCS)	Case 2: CUEA Student Portal
2	Transcript request and upload module integrated within the clearance workflow	Transcript handling not integrated within the clearance module
3	Digital clearance certificate generated and available for download on completion	Clearance certificate not generated digitally; collected through a separate process
4	Sequential workflow with defined approval order enforced by the system	Departmental clearance updates managed independently without enforced sequence
5	Designed specifically for the Tangaza University clearance structure and roles	Generalised institutional portal not tailored to a specific clearance workflow

2.3.3 Case 3: Strathmore University Clearance System

Strathmore University developed an internal clearance module accessible through its main student portal. Students can submit clearance requests online and receive departmental feedback without visiting offices in person. The system maintains a digital archive of all clearance records, which has improved record management considerably and enabled administrators to retrieve historical data when needed (Strathmore ICT Report, 2021).

The system's user interface was reported to be intuitive and contributed to a relatively smooth adoption by staff. However, the system does not send reminder notifications to students who have pending clearance steps, which means students occasionally miss graduation deadlines due to inaction. The system was also developed as a closed, proprietary product tightly integrated with Strathmore's own infrastructure, making it difficult to adapt for other institutions. The proposed SGCS improves on this by incorporating automated reminder notifications and being built entirely on widely documented standard technologies.

Table 3: Comparison of Proposed System with Case 3 (Strathmore University)

S/N	Proposed System (SGCS)	Case 3: Strathmore System
1	Automated reminder notifications for pending clearance items	No automated reminders for students with pending items
2	Built on standard, widely documented technologies	Proprietary, closed system
3	Escalation notification if a department delays approval	No escalation mechanism for delayed approvals
4	Digital archive of all clearance records	Digital archiving available
5	Adaptable to other institutions with minimal modification	Not adaptable to other institutions

2.3.4 Case 4: Moi University Student Clearance System

Moi University developed a student clearance module within its Student Information System. The module's most notable feature is its direct integration with the fee management system: when a student has an outstanding fee balance, the system automatically blocks clearance progress until the balance is resolved. This eliminates the need for manual fee verification and significantly reduces the risk of a student obtaining clearance while still owing money to the university.

The proposed SGCS replicates the intent of this approach through a financial clearance verification step handled by the Finance Officer, who can flag an outstanding balance to block progression until it is resolved. However, Ruto and Sang (2022) found that the Moi University system performs poorly during peak usage periods near graduation, with noticeable slowdowns when many students are processing clearance simultaneously. The system also lacks a mobile-optimised interface. The proposed SGCS addresses the usability concern through a responsive web design accessible across device types.

Table 4: Comparison of Proposed System with Case 4 (Moi University)

S/N	Proposed System (SGCS)	Case 4: Moi University SIS
1	Mobile-responsive interface accessible on smartphones	No mobile-responsive interface
2	Designed for reliable performance under concurrent student load	Slowdowns reported during peak graduation season
3	Finance Officer verification step to block clearance for fee arrears	Automated fee blocking integrated with fee system
4	Reporting dashboard for Director and Registrar	Limited reporting features
5	Standard server-side scripting without complex infrastructure	Complex server infrastructure required

2.3.5 Case 5: JKUAT Online Clearance Module

JKUAT integrated a clearance module into its Student Online Services Portal. Students can upload scanned supporting documents as part of their clearance submission, and officers can review these documents online and either approve or request additional information without requiring the student to visit in person. This feature has meaningfully reduced the volume of student visits to JKUAT's administrative offices (Ngugi, 2021).

However, the system has two notable limitations. First, there is no escalation mechanism: if a departmental officer fails to act on a clearance request within a reasonable time, neither the

student nor a supervisor receives an automatic notification, and delays can extend indefinitely. Second, even after all departmental approvals are obtained, students are required to visit the registrar's office physically to collect the official clearance document. The proposed SGCS addresses both limitations by incorporating automated escalation notifications and enabling digital clearance certificate generation for download within the system.

Table 5: Comparison of Proposed System with Case 5 (JKUAT)

S/N	Proposed System (SGCS)	Case 5: JKUAT Module
1	Automated escalation notification if a department delays beyond a defined period	No escalation mechanism for delayed departmental approvals
2	Digital clearance certificate generated and available for download within the system	Student must collect physical certificate at the registrar's office
3	Supporting document upload for officer verification	Document upload available
4	Real-time approval status visible to student per department	Status display is limited
5	Full audit trail maintained in the centralised database	Basic audit trail only

2.4 Emerging Trends and Patterns

A review of the literature and the five related systems reveals several important trends that are shaping the development of student information and clearance systems in higher education. Each trend is observable not only in the broader literature but in the specific systems reviewed in this chapter, and each has directly informed design decisions in the proposed SGCS.

The first and most consistently observed trend is the shift towards mobile-first design. Smartphone penetration in Kenya exceeded sixty percent in 2023 (Communications Authority of Kenya, 2023), and students overwhelmingly access online services through mobile devices. Of the systems reviewed, this gap is most visible at Moi University, whose clearance module was found by Ruto and Sang (2022) to have no mobile-responsive interface, contributing to poor usability among students accessing the system via smartphone. Similarly, the University of Nairobi clearance portal was identified by Omondi (2020) as lacking mobile responsiveness, a significant limitation given the device preferences of Kenyan students. By contrast, the JKUAT Online Clearance Module has made progress in this area by ensuring its student-facing interface functions across device types, reflecting an institutional recognition that mobile accessibility is no longer optional. The proposed SGCS will adopt a responsive web design

approach to ensure full usability across screen sizes, directly addressing the gap observed in the UoN and Moi University systems.

A second significant trend is the increasing reliance on cloud-based and centrally hosted systems that enable location-independent access to administrative services. Rather than requiring students and officers to be physically present at an office or connected to a campus network, cloud-hosted portals allow interaction with institutional systems from any internet-connected device. The CUEA student portal exemplifies this trend: by hosting its clearance and academic services on a central web platform, CUEA allows students to access their records and clearance status from off-campus. The KUCCPS placement portal, which manages student placement for thousands of concurrent users nationally, further demonstrates the scalability that cloud-based infrastructure makes possible, having maintained high availability even during peak application periods (KUCCPS, 2022). The proposed SGCS will be designed as a centrally hosted web-based system consistent with this approach, removing the need for students at Tangaza University to visit offices in order to check or advance their clearance status.

Third, the automation of administrative workflows is becoming standard practice in well-functioning university systems. Rather than relying on manual coordination between departments, modern systems progress the clearance process automatically: notifying the next department when a prior step is completed, alerting students when action is required, and escalating unresolved requests after a defined period. The Strathmore University clearance system illustrates this trend in part, it allows officers to review and update clearance records online without requiring in-person visits, but falls short in that it does not send automated reminder notifications to students with pending items, a gap that Strathmore ICT Report (2021) acknowledged has led students to miss graduation deadlines. The JKUAT module goes further by allowing document uploads and officer review online, but similarly lacks an escalation mechanism, meaning delays can persist indefinitely without automatic follow-up (Ngugi, 2021). The proposed SGCS will implement the full automated workflow that these systems approach but do not complete, including sequential departmental progression, per-stage student notifications, and escalation alerts for delayed approvals.

Fourth, the generation of digital certificates and electronic transcripts is an increasingly important feature of academic administration systems. Digital documents can be issued instantly upon completion of all requirements, are more resistant to forgery than paper equivalents, and eliminate the need for students to make an additional physical visit to collect

printed materials. Among the reviewed systems, this gap is most apparent at JKUAT, where Ngugi (2021) found that students who have completed all departmental approvals online are still required to collect their official clearance document physically from the registrar's office, undermining the convenience the digital workflow was intended to provide. The University of Nairobi system similarly issues printed certificates at the registrar's office rather than generating them digitally (Omondi, 2020). The CUEA portal, while it includes a clearance module, also does not generate certificates digitally for immediate download. The proposed SGCS will address this gap directly by generating a digital clearance certificate within the system upon final approval by the Registrar, available for immediate download without any further physical visit.

Finally, a growing body of research and institutional experimentation is exploring the use of artificial intelligence and automated response systems within university administrative portals. Chatbots and AI-assisted query handling are beginning to appear in student-facing systems to provide immediate responses to common questions about clearance status, required documentation, and outstanding obligations, reducing the volume of routine enquiries that staff must handle manually. None of the five systems reviewed in this chapter has yet implemented AI-assisted features in their clearance or student services workflows, reflecting the early stage of this trend in the Kenyan higher education context. Globally, institutions such as the Massachusetts Institute of Technology have experimented with blockchain-based digital credentials (MIT Media Lab, 2018), and AI-driven student service tools are increasingly documented in North American and European university literature (Adeyemi & Salawu, 2022). While the proposed SGCS will not implement AI features in its initial version, awareness of this trajectory has informed the system's design principle of reducing manual staff intervention at every stage of the clearance workflow, laying a foundation upon which intelligent features could be added in a future iteration.

2.5 Chapter Summary

This chapter reviewed five related systems; the University of Nairobi Clearance Portal, the Catholic University of Eastern Africa Student Portal, the Strathmore University Clearance System, the Moi University Student Information System, and the JKUAT Online Clearance Module; identifying the strengths and weaknesses of each and comparing them against the proposed SGCS. The historical development of student administrative systems in higher education was traced from the text-based systems of the 1980s through to today's integrated, web-based platforms. Five emerging trends were identified: mobile-first design, cloud-based

centralisation, automated workflows, digital certificates, and AI-assisted services. Across the reviewed systems, common gaps include the absence of mobile responsiveness, limited or absent per-event notification, no escalation mechanism for delayed approvals, and the requirement for physical certificate collection. The proposed SGCS is designed to address each of these limitations within a single, cohesive system built on standard, widely supported web technologies. Chapter Three now describes the methodology that will guide the development of the proposed system.

CHAPTER THREE: METHODOLOGY

3.1 Introduction

This chapter describes the approach and methods that will be employed to research, design, build, test, and deploy the Students' Graduation Clearance System. The methodology chapter serves an important function in any systems development project: it makes explicit the decisions taken about how the work will be conducted, and it justifies those decisions in terms of the nature of the system being built and the context in which it will operate. Each section of this chapter addresses a distinct phase of the development process, from the initial gathering of requirements through to final deployment. For every concept introduced, a definition is provided, its role in the project is explained, and the reasons for its selection are justified. All sections are written in the future tense, as the development work will commence upon approval of this proposal.

3.2 Research Approach

A research approach is a structured plan that defines how a study is conducted, guiding the researcher from the formulation of the problem through to the delivery of a solution. In the context of information systems development, the research approach determines the sequence and nature of activities undertaken: what information is collected and how, how the existing system is studied, how the proposed system is designed, and how the finished product is evaluated. A clearly defined approach ensures that the project proceeds in a logical and organised manner, that no critical phase is overlooked, and that the methods chosen are appropriate to the goals of the study.

The Structured System Analysis and Design (SSAD) methodology has been selected to guide the development of the SGCS. SSAD is a well-established, process-oriented approach to information systems development that provides a clearly defined sequence of phases, requirement elicitation, system analysis, design, implementation, testing, and deployment, each supported by standardized tools and techniques. It has been chosen for this project for three principal reasons. First, the graduation clearance process at Tangaza University is well-defined and rule-based, with a consistent sequential structure that maps naturally to SSAD's phase-by-phase framework. Second, SSAD's use of graphical tools such as context diagrams and data flow diagrams makes it straightforward to represent and discuss the system with non-technical stakeholders, including the administrative staff who will ultimately use it. Third, as a methodology with a long and documented history of application in academic and institutional

information systems development, SSAD provides a principled and defensible basis for a project of this nature.

3.3 Requirement Elicitation

Requirement elicitation is the process of identifying, in precise and practical terms, what the proposed system must do, as determined by the people who will use it. It is the foundation of the entire development process: a system that is designed without accurate requirements is likely to fail to meet user needs regardless of the quality of its technical implementation. For the SGCS, two complementary data collection methods will be used: document analysis of official Tangaza University administrative forms and questionnaires administered to final-year students. Each method will be suited to a different type of information and a different source of data.

3.3.1 Document Analysis

Document analysis is a systematic research method in which existing written, printed, or official records are examined and interpreted to extract information relevant to the study. Unlike interviews or questionnaires, document analysis draws on materials that were produced independently of the research process, meaning they reflect actual institutional practice rather than self-reported recollections. For the purposes of requirement elicitation, this method involves collecting, reading, and analysing official documents that govern or record the process under investigation.

For this study, two official Tangaza University documents will be obtained and analysed to establish a clear, evidence-based understanding of the current graduation clearance process. The first document is the Tangaza University Application for Graduation form, used for the 2025 graduation cycle. This form is completed by students who have finished their programme and wish to apply for graduation. It collects the student's registration number, contact information including postal address, date of birth, phone number, and email address, gender, national identification or passport number, institute and programme, award type, specialisation where applicable, graduation date, and the full name as it should appear on official documents. It also includes a Registration and Course Information section in which credits for core courses, exempted credits, and the semester and academic year of programme completion are recorded. The form includes a Declaration by the Student section and, critically for the proposed system, an Academic Programme Leader section in which the programme leader confirms whether the student is eligible to graduate, provides comments, and signs and stamps the form before

forwarding it to the Registrar's Office. Analysis of this document will inform the student registration data model for the SGCS, confirming the fields that must be stored in the student profile, and established that eligibility confirmation by the academic programme leader is a prerequisite step that occurs before the main departmental clearance workflow begins.

The second document to be analysed is the Tangaza University Student Clearance Form², which is the physical paper instrument currently used to process graduation clearance at the institution. This form is divided into six sections, each corresponding to a department that must verify and sign off on the student's clearance. The sections, in the order they appear on the form, are as follows: Institute Clearance, in which the Director of the student's institute confirms whether all university materials have been returned or whether materials belonging to the university remain outstanding, recording any comments and providing a signature and stamp; Library Clearance, in which the Librarian confirms whether all library books and materials, including library cards, have been returned, or whether library books and materials of a stated value remain outstanding; Office of the Dean of Students, in which the Dean confirms the student's conduct standing, with the same binary outcome of clearance or outstanding material noted; Administration/University Store, in which the Administration or Store Officer confirms whether all materials belonging to the university have been returned or whether items of a stated value in Kenyan Shillings remain with the student; Finance Office, in which the Finance Officer confirms whether the student has no outstanding debts or arrears, or records the total amount due across all departments and institutes, noting the date of clearance; and the Registrar's Office, which provides the final confirmation that the student has cleared with all departments and institutes and has no outstanding balance, or notes that clearance is yet to be completed. Each section contains a comments field and a space for the authorised officer's signature and stamp. Analysis of this document will establish the precise sequential workflow that the proposed SGCS must replicate digitally, identify the data fields required at each stage, and confirm the binary decision logic, cleared or not yet cleared, that governs each departmental step.

Document analysis will serve as the most appropriate primary method for establishing the workflow requirements of this system because the documents to be analysed are the actual artefacts currently used to manage the process. They'll provide an authoritative, unambiguous record of what each department checks, what information is recorded at each step, and in what sequence approvals are obtained. This eliminates the ambiguity that can arise when relying solely on participants' verbal accounts of a process, which may vary between individuals or

reflect ideal rather than actual practice. The findings from document analysis will directly inform the database design, the approval workflow logic, and the data fields implemented in the proposed system.

3.3.2 Questionnaires

A questionnaire is a structured set of written questions distributed to a group of respondents for the purpose of collecting data in a standardised and efficient format. Because all respondents answer the same questions in the same form, the responses can be compared, counted, and analysed consistently across the group. Questionnaires are particularly well-suited to gathering information from larger populations where individual interviews would be impractical.

Questionnaires will be administered to forty final-year students selected through simple random sampling from Tangaza University's final-year student population. The instrument will include both closed-ended and open-ended questions and has been designed to take no more than ten minutes to complete. Questions will address students' direct experiences with the current clearance process, the challenges they have encountered, and the features they would consider most valuable in an online system. Questionnaires will be distributed and filled out as a google form. Responses will be analysed using frequency tables and percentage calculations, with key findings presented in charts where this aids clarity.

Questionnaires will be the appropriate method for this group because the questions to be asked are sufficiently straightforward to be answered in writing without requiring live follow-up, and because collecting equivalent data from forty students through individual interviews would not be feasible within the project timeline. The method will also ensure that the design of the system reflects a broad cross-section of student perspectives rather than the experience of a small number of individuals.

3.4 System Analysis

System analysis is the systematic process of studying and documenting how an existing system functions in order to understand it thoroughly enough to design an improved alternative. It involves identifying the processes that currently take place, the data that flows between them, the roles of the people involved at each stage, and the points at which the current system fails to meet the needs of its users. In the context of the SGCS, this means examining the current manual graduation clearance process at Tangaza University before any design decisions are

made, because a proposed system designed without this understanding is likely to replicate the very inefficiencies it is intended to solve.

The SSAD methodology will guide the analysis phase, using standardised graphical tools to document both the current and proposed systems. A Context Diagram will provide a high-level representation of the system's scope, showing all external entities, students, departmental officers, and the Academic Registrar, and the data flows between them and the central process. This establishes the boundaries of the system clearly. A Level 1 Data Flow Diagram (DFD) will decompose the major processes within the system into greater detail, showing how data moves through each step, where it is stored, and where delays or failures are most likely to occur under the current manual arrangement. These tools will be applied to both the current system and the proposed SGCS, so that the differences between them are visible and documented. Database normalisation up to Third Normal Form (3NF) will also be applied to the data structures identified during analysis, ensuring that the database design is logically consistent and free of unnecessary redundancy before design work begins.

System analysis will serve as an indispensable step in the development process because it will ground every subsequent design decision in an accurate understanding of the problem being solved. Proceeding directly from a problem statement to a design solution without rigorously studying the existing system is a well-documented source of project failure in information systems development, and this analysis phase will ensure that the proposed system is built to address real needs rather than assumed ones.

3.5 System Design

System design is the process of translating the requirements and understanding produced during analysis into a detailed, actionable plan for how the proposed system will be built. Where analysis answers the question of what the system must do, design answers the question of how it will do it, specifying the structure of the database, the flow of data and control through the application, and the way in which users will interact with the system at each stage. Investing sufficient effort in design before coding begins reduces the likelihood of discovering fundamental structural problems during implementation, and produces a system that is easier to understand, maintain, and extend.

The database design for the SGCS will be defined using an Entity Relationship Diagram (ERD), which identifies the key entities in the system, such as the student, the clearance request, the department, and the officer, along with the relationships between them and the

attributes that each entity must store. The database will be normalised to Third Normal Form to eliminate redundancy and ensure data integrity. The process flow design will be documented through a Level 2 DFD that expands the clearance initiation process to the sub-process level, and through a system workflow flowchart that maps the complete clearance journey from student login to certificate download, including all decision points at which a department may approve or reject a request. User interface design will be developed through wireframe prototypes of the key screens, including the student clearance dashboard and the departmental officer approval interface. These prototypes allow the design to be reviewed and refined with stakeholders before any code is written, serving both as a communication tool and as a reference for frontend development.

Design will be a necessary and valuable phase of the project because errors caught at the design stage will cost significantly less to correct than those discovered during or after implementation. A well-executed design will also make the development process more efficient by providing clear direction and reducing the need for improvisation during coding.

3.6 System Implementation

System implementation is the phase in which the design is translated into working software through the actual writing of code. The technology stack for the SGCS will be selected according to a clear criterion: each tool must be a standard, well-documented technology that is widely understood, maintainable by staff with general web development skills, and appropriate to the scale and requirements of a single-institution clearance system.

3.6.1 Frontend Development

The frontend of a web application is the client-side layer that users interact with directly through their web browser. It is responsible for presenting information clearly, collecting user input, and communicating with the server-side application in response to user actions. The quality of the frontend has a direct impact on whether the system is used effectively: a poorly designed interface creates friction for users regardless of how correctly the underlying logic operates.

The SGCS frontend will be developed using HTML to define the structure and content of each page, CSS to control the visual presentation and layout, and JavaScript to provide dynamic behaviour such as form validation, status updates, and interactive elements. Together, these three technologies constitute the standard foundation of web interface development and are supported by all modern browsers without additional plugins or installations. The interface will

be designed to be responsive across different screen sizes, ensuring that students and officers can use the system effectively from a desktop computer, a tablet, or a smartphone. This is a deliberate design priority, given that the majority of students in the target user group access online services through mobile devices.

Frontend development is important because it is the layer of the system that users experience directly. A well-structured, accessible interface reduces the time users need to learn the system, minimises errors in data entry, and encourages consistent adoption, all of which are essential to the system achieving its intended impact.

3.6.2 Backend Development

The backend of a web application is the server-side layer responsible for processing requests received from the browser, executing the business logic of the application, enforcing access control, and coordinating the flow of data between the user interface and the database. In the context of the SGCS, the backend is where the clearance workflow is enforced: it determines which department is currently active in the sequence for a given student, validates approval and rejection actions submitted by officers, triggers notifications at the appropriate points in the process, and controls what each user type is permitted to see and do within the system.

PHP has been selected as the server-side scripting language for the SGCS backend. PHP is a widely used, well-documented language specifically designed for web development, with a large body of available reference material and strong community support. It runs on standard web servers and integrates naturally with relational databases, making it well-suited to the data-driven, workflow-based requirements of the SGCS. PHP is also broadly familiar to web developers, which ensures that the system can be maintained and updated by institutional technical staff without dependence on specialised expertise.

Backend development is important because it is where the system's core logic resides. The correctness and security of the backend determines whether the clearance workflow operates as intended, whether user data is handled appropriately, and whether the system behaves consistently under normal and unexpected conditions.

3.6.3 Server-Side and Database Management

Server-side processing refers to the computation that takes place on the web server in response to a user request, before a response is returned to the client browser. Database management refers to the organised storage, retrieval, updating, and integrity enforcement of the application's data. Together, these two concerns form the foundation of the system's ability to

persist information reliably across sessions, enforce business rules at the data level, and support the multi-user, sequential workflow that the clearance process requires.

A relational database(MySQL) will be used to store all system data, including student records, clearance requests, departmental approval statuses, officer actions, and notification logs. A relational database is structured around tables with defined relationships between them, governed by a schema that enforces data types, constraints, and referential integrity rules. This structure ensures that data is stored consistently, that invalid states cannot arise through programming errors or user input, and that queries can retrieve information accurately and efficiently. The database will be designed in accordance with the ERD produced during the system design phase and will be normalised to Third Normal Form to eliminate redundant data and ensure update consistency.

The use of a relational database will be justified by the nature of the clearance process itself: each clearance request will be related to a specific student, a specific set of departments, and a specific series of officer actions, and these relationships will need to be maintained accurately for the system to function correctly. A relational database will provide the structural guarantees necessary to manage these relationships at scale without data integrity failures.

3.7 System Testing

System testing is the process of verifying that a completed system functions correctly, performs reliably under realistic conditions, and meets the needs of the users for whom it was designed. Testing is not a single activity conducted at the end of development, but a layered and progressive process that moves from technical validation of individual components to practical evaluation of the system as a whole. For the SGCS, testing will be conducted in three stages.

The first stage is developer testing, conducted by the researcher throughout the implementation phase. This involves unit testing of individual components, verifying that each function, database query, and page handler behaves correctly in isolation, and integration testing to confirm that the components work together as expected. For example, tests will confirm that approving a clearance step correctly advances the workflow to the next department, that rejection records the officer's comment and notifies the student, and that the system prevents a student from proceeding to the Dean's approval while a Finance Office flag remains unresolved. Defects identified during developer testing will be corrected before the system is presented to external users.

The second stage is end-user testing, also referred to as User Acceptance Testing (UAT). A group of five final-year students and two administrative staff from Tangaza University's clearance departments will be invited to use a deployed test version of the system and complete a set of predefined tasks, including initiating a clearance request as a student, reviewing and approving a request as a departmental officer, and checking clearance status from the student dashboard. Feedback will be collected using a short structured evaluation form and incorporated into the system through a final round of revisions before the system goes live. The UAT session will be held on campus over a single afternoon.

The third stage is heuristic evaluation, in which the system's interface will be assessed against established usability principles, such as visibility of system status, error prevention, and consistency, to identify usability issues that may not surface through task-based testing. Findings from this evaluation will inform final interface adjustments before deployment.

Testing will be essential to the success of the project because errors in a clearance workflow will have real consequences for the students and officers who depend on it. Thorough testing will be what makes it possible to deploy the system with confidence that it will behave as intended.

3.8 System Deployment

System deployment is the process of making a completed and tested system available for operational use within the institutional environment for which it was designed. Deployment marks the transition from a development artefact to a working tool that real users rely on for real tasks. In the case of the SGCS, this transition involves moving from a development and testing environment to a live, accessible web application that Tangaza University students and staff can use to manage the graduation clearance process.

The deployment of the SGCS will follow a phased approach. In Phase 1, a pilot deployment will be conducted with one department, the Library, and a small group of five students, using the live system to process actual clearance requests. This pilot phase is designed to identify any issues that did not emerge during testing under controlled conditions, and to provide an opportunity for correction before the system is extended to all users. Issues identified during the pilot will be resolved before Phase 2 begins. In Phase 2, all six clearance departments and all eligible final-year students will be given access to the system, and it will become the primary channel for graduation clearance at Tangaza University.

The transition from the current manual process to the digital system will be managed carefully to avoid disruption. During the pilot phase, the manual paper-based process will remain available as a fallback. Training will be provided for departmental officers prior to both phases of deployment, covering how to log in, view pending clearance requests, submit approvals or rejections, and interpret the reporting features available to the Director and Academic Registrar. A printed one-page quick reference guide will be distributed to officers, and a help section will be available within the system for students. A data migration plan will be established to ensure that any student records that need to be initialised in the system before go-live are entered accurately and completely.

Careful deployment planning will be important because even a technically sound system can fail to achieve adoption if it is introduced without adequate preparation, training, and transition support. The phased approach, combined with structured training and a defined fallback, will give the system the best possible conditions for successful institutional uptake.

3.9 Chapter Summary

This chapter has described the complete methodology that will guide the development of the Students' Graduation Clearance System. Requirements will be gathered through document analysis and questionnaires administered to thirty final-year students. System analysis and design will be conducted under the SSAD methodology, producing context diagrams, Level 1 and Level 2 DFDs, a system workflow flowchart, an Entity Relationship Diagram, and user interface prototypes. The implementation will use HTML, CSS, and JavaScript for the client-side interface and PHP for server-side scripting, supported by a relational database normalised to Third Normal Form. Testing will progress through developer unit and integration testing, User Acceptance Testing with real Tangaza University users, and heuristic evaluation of the interface. Deployment will follow a phased approach, pilot with one department, then full rollout, with structured training for officers and a defined transition plan.

CHAPTER FOUR: SYSTEM ANALYSIS

4.1 Introduction

System analysis is the disciplined process of investigating and understanding how an existing system operates so that a superior replacement can be designed with precision and purpose. It is not merely an academic formality but a practical necessity: a system designed without a rigorous analysis of the environment it is meant to serve risks perpetuating the same inefficiencies it was intended to eliminate, or introducing new problems by failing to account for constraints that are only visible once the existing workflow is properly understood. In the context of this project, system analysis represents the bridge between the research grounding established in Chapters One through Three and the design decisions that will be presented in Chapter Five.

Chapter Three described the SSAD methodology that governs this project and identified system analysis as one of its core phases. That chapter explained that analysis would involve the use of context diagrams and data flow diagrams to document both the current and proposed systems, and that the findings from document analysis and student questionnaires would provide the empirical foundation for this phase. The present chapter fulfils that commitment. It begins by describing the current manual graduation clearance process at Tangaza University in sufficient detail to illuminate not only how it functions in theory but how it operates in practice, including the specific points at which it fails students and administrative staff. It then examines the genuine strengths of the current system before undertaking a comprehensive assessment of its weaknesses. A full feasibility study evaluates whether the proposed Students' Graduation Clearance System (SGCS) is achievable, affordable, and operationally sensible. The chapter then specifies the functional and non-functional requirements of the proposed system, and documents the principal analysis artefacts produced under the SSAD methodology, including context diagrams, data flow diagrams, system flowcharts, and actor descriptions.

4.2 Description of the Current System

The graduation clearance process at Tangaza University, as it currently exists, is conducted entirely through the circulation of a single physical paper form. When a student believes they have satisfied all academic and financial requirements for graduation, they are required to obtain this form from the Office of the Academic Registrar and proceed, office by office, through six separate departments, each of which must verify compliance from its own

perspective and append a signature and institutional stamp before the student can advance to the next stage.

The process begins with the student visiting the Institute or Department Directorate. The Director, or a designated delegate, reviews whether the student has returned all materials belonging to the university that were issued through the institute. This may include items such as laboratory equipment, academic resources, or departmental property. If everything is in order, the Director signs and stamps the clearance form under the Institute Clearance section. If items are outstanding, the value is noted in Kenyan Shillings and the form cannot proceed until the matter is resolved. Having secured the Director's signature, the student then travels to the University Library, where the Librarian must confirm that all borrowed books and library materials, including the student's library card, have been returned. Outstanding library books or materials are similarly noted with their monetary value. The form moves on only after the Librarian is satisfied.

The third stop is the Office of the Dean of Students, where the student's conduct record and any outstanding obligations relating to student affairs are reviewed. The Dean or an authorised officer confirms clearance or notes any materials or conduct issues requiring resolution. From the Dean's Office, the student proceeds to the Administration and University Store, where the store officer verifies that all items issued by the institution have been returned. The Finance Office is typically the most consequential station in the process: the Finance Officer must confirm that the student has no outstanding fees, debts, or arrears. If a balance exists, its total amount is recorded, and the student is effectively blocked from completing clearance until payment is made. The final stop is the Office of the Academic Registrar, which provides the definitive confirmation that the student has successfully cleared with all departments and has no outstanding financial obligation. The Registrar's signature and stamp on the clearance form signals that the student is administratively eligible for graduation.

In practice, this sequence is rarely completed in a single day. Officers are not always present when students arrive: a departmental meeting, a busy period at the library, or an officer attending to other institutional responsibilities can mean that a student arrives at an office only to find no one available to process their form. There is no appointment system, no ticketing mechanism, and no way for a student to verify in advance whether an officer is available. Students frequently return to the same office two, three, or even more times before securing a signature. Because there is no digital record of where each student stands in the process, the

only way for a student to determine their clearance status is to physically visit the relevant office. Transportation costs, time, and examination commitments compound the stress of the process during what is already a demanding period of the academic year.

Communication between departments is minimal and informal. If a student has an outstanding library fine that the Finance Office is not yet aware of, or if an issue arises at the Dean's Office that requires the Director's input, communication happens through telephone calls or in-person conversations between officers, introducing further delays. There is no mechanism for a student to receive proactive notification that a problem has been identified and needs their attention. Students learn of problems by arriving at an office and being turned away, or by discovering at the Finance Office that an obligation they believed to have been settled has not been formally recorded. In cases where the clearance form is lost, a scenario that the document analysis confirmed does occur, the process must begin again from the start, with the student returning to every department to obtain fresh signatures.

4.3 Strengths of the Current System

Despite its limitations, the current manual clearance process at Tangaza University possesses a number of genuine strengths that informed the design of the proposed SGCS and that account for why the system has continued to function in the absence of a digital alternative.

The most significant strength of the manual system is its conceptual simplicity. The process is straightforward to understand: a student obtains a form, visits a series of offices, collects signatures, and returns the completed form to the Registrar. There are no login credentials to remember, no interfaces to navigate, and no technical failures to contend with. For students and officers who are not highly comfortable with digital tools, this simplicity reduces the barrier to participation. Every person involved in the process has a clear and immediate understanding of what they are expected to do.

A related strength is the low infrastructure requirement. The current system operates entirely on paper and requires no ICT investment, no network connectivity, no servers, and no software maintenance. This is particularly relevant in an institutional context where resources are constrained and the technical support capacity may be limited. The system can continue to function during power outages, internet disruptions, or hardware failures that would disable a digital alternative. Its operational continuity under adverse conditions is a real, if often unacknowledged, asset.

Physical verification is another area where the manual system has an inherent advantage. When a Librarian physically checks a student's record against the library database and hands over the clearance stamp in person, the interaction involves direct human judgement and the application of institutional knowledge that is difficult to replicate fully in a digital workflow. An officer who knows a student personally, or who has specific institutional context about an outstanding obligation, can apply discretion in ways that a rule-based automated system cannot. The face-to-face interaction also provides an opportunity to resolve minor ambiguities on the spot in a way that an asynchronous digital approval cannot easily replicate.

Finally, the manual system is familiar to all current users. Staff have been processing clearance forms for years and understand the expectations, the workflow, and the informal norms that govern the process. This familiarity reduces the need for training and eliminates the risk of user confusion or resistance that accompanies the introduction of any new system. From a strictly operational perspective, the current system works for those who have always used it, and its familiarity is a genuine, if intangible, asset that any replacement system must work to replicate through good interface design and adequate user training.

4.4 Weaknesses of the Current System

The weaknesses of the current manual clearance process at Tangaza University are substantial and compound one another in ways that have a measurable negative impact on both students and administrative staff. Each weakness identified below was corroborated through document analysis and questionnaire responses collected from final-year students, and each has directly shaped the requirements specification for the proposed SGCS.

Long queues and repeated visits are the most immediately visible problem. Because all students eligible for a given graduation cycle must pass through the same six offices within a compressed timeframe, the period immediately before graduation creates significant congestion at each department. Officers are occupied for extended periods processing individual students, and those waiting have no way of knowing how long the wait will be or whether they could return at a more convenient time. Students reported in questionnaire responses that it was common to make three or more separate visits to a single office before securing clearance, a pattern that imposes significant costs in time, transport, and personal energy.

The complete absence of real-time tracking is a pervasive and structurally significant weakness. Under the current system, a student has no way of knowing, without visiting the campus in

person, which departments have cleared them and which have not. There is no digital record, no online portal, and no notification mechanism. The information a student needs to manage their own clearance process is held entirely within a piece of paper that passes between offices and is inaccessible from any distance. This creates particular hardship for students who live far from campus, who are completing work-based learning obligations, or who are sitting examinations during the clearance period.

The risk of lost or damaged forms is a direct consequence of relying on physical paper as the sole record of clearance progress. The Student Clearance Form reviewed during document analysis is a single physical document that travels between six offices over an extended period. It can be misplaced at any point in the process, left behind in an office, damaged by water or wear, or simply forgotten in a pile of papers. When a form is lost, the student must restart the entire process, visiting each department again to obtain fresh signatures. This is not merely an inconvenience but a significant institutional failure that reflects the absence of any backup or audit mechanism in the current system.

Poor inter-departmental communication compounds the delays described above. If the Finance Office needs to query a record held at the Director's Office, or if the Registrar has a question about a student's conduct record from the Dean's Office, the only available channels are telephone calls or in-person visits between officers. There is no shared information system through which departments can communicate about a student's case, view each other's records, or flag outstanding issues for resolution. Each department operates in relative isolation, unaware of what other departments have or have not processed.

Data redundancy and integrity risks arise because the same student information is recorded repeatedly at each department. Each officer fills in the student's name and registration number on the clearance form, and each department may hold separate records of the same student in its own filing system. This creates multiple opportunities for inconsistencies to arise, whether through transcription errors, variations in how a student's name is recorded, or discrepancies between what is on the clearance form and what a department's own records show. There is no mechanism for cross-referencing these records automatically, meaning errors can persist undetected.

From an administrative perspective, the lack of any centralised reporting capability is a serious limitation. The Academic Registrar has no reliable way to determine, at any given moment, how many students have completed clearance, how many are still in progress, which

departments are creating bottlenecks, or whether the institution is on track to clear all eligible students before the graduation ceremony. Generating such information under the current system would require manually collecting and aggregating information from six separate offices, a task that is both time-consuming and prone to error. This absence of institutional oversight makes it difficult to manage the graduation clearance process strategically and undermines the institution's ability to respond to delays before they become critical.

Finally, the human cost of the current system, particularly the stress and anxiety experienced by students, should not be underestimated. Kariuki (2020) established that the stress associated with manual clearance processes has a measurable negative impact on student wellbeing during an already demanding period of the academic year. The combination of uncertainty about clearance status, repeated journeys to campus, the risk of missing a deadline because an officer was unavailable, and the knowledge that a single lost form can undo weeks of effort creates an experience that is needlessly difficult and that reflects poorly on the institution's commitment to student welfare.

4.5 Feasibility Study

Before committing to the development of any information system, it is essential to assess whether the proposed solution is genuinely achievable, affordable, and operationally viable within the specific environment in which it is intended to function. The feasibility study presented here examines the proposed SGCS from five distinct perspectives: technical, economic, operational, schedule-related, and legal and ethical. Each dimension is evaluated independently, and the section concludes with a consolidated finding.

4.5.1 Technical Feasibility

Technical feasibility addresses the question of whether the proposed system can be built using currently available technologies, and whether the operating environment at Tangaza University can support it in practice. The answer to both questions is clearly affirmative. The SGCS is designed using a technology stack consisting of HTML, CSS, and JavaScript for the client-side interface, PHP for server-side scripting, and MySQL for relational database management. Each of these technologies is mature, widely adopted, and extensively documented. PHP in particular was designed specifically for web application development and integrates seamlessly with MySQL, making it the natural choice for a data-driven, workflow-based system of this type. The SGCS does not require any specialised hardware, proprietary platforms, or expensive commercial software licences.

The system will be hosted on a standard shared web hosting environment, which is readily available from Kenyan hosting providers at a modest cost and provides the web server, PHP runtime, and MySQL database server required to run the application. Internet penetration in Kenya is well-established, and the Communications Authority of Kenya (2023) reported smartphone penetration exceeding sixty percent nationally, with connectivity widely available in Nairobi, where Tangaza University is located. Students accessing the system from off-campus via mobile devices or home internet connections will be able to do so without any technical barriers.

Administrative staff at Tangaza University already use computers as part of their daily work, and general ICT literacy at the institution is sufficient to support the adoption of a web-based system. No advanced technical skills are required to operate the SGCS as an end user: the ability to navigate a web browser and interact with straightforward form-based interfaces is the only prerequisite. The technical feasibility of the SGCS is therefore well established.

4.5.2 Economic Feasibility

Economic feasibility considers whether the cost of developing and operating the proposed system is justified by the benefits it delivers, and whether the financial commitment is within the means of the institution. The development costs for the SGCS are modest, primarily because the entire technology stack is open-source and freely available. PHP, MySQL, XAMPP for local development, Visual Studio Code as the development environment, and Git for version control all carry no software licensing costs. The principal financial outlays are web hosting, domain registration, internet subscription during development, and documentation costs, which together amount to an estimated KES 22,110 as detailed in Appendix C of the proposal.

When considered against the long-term operational costs of the current manual system, the economic case for the SGCS is compelling. The paper-based process imposes direct costs through stationery, printing, and physical filing, as well as indirect costs through the staff time consumed by manual coordination, the repeated processing required when forms are lost, and the administrative overhead of responding to student enquiries about clearance status. A system that handles these tasks automatically at negligible marginal cost per student represents a significant long-term saving. Over the course of successive graduation cycles, the efficiency gains will substantially outweigh the modest one-off development investment.

4.5.3 Operational Feasibility

Operational feasibility examines whether the proposed system will be accepted and used effectively by the people it is designed to serve. For the SGCS, two primary user groups must be considered: the final-year students who will use the system to initiate and track their clearance, and the departmental officers who will use it to process approval requests.

Student acceptance is expected to be high. The questionnaire administered to forty final-year students during the requirement elicitation phase confirmed that the overwhelming majority would prefer an online clearance system to the current manual process. Students identified real-time status tracking, per-department notifications, and the ability to initiate clearance from any location as the features they would value most. These are precisely the features that the SGCS delivers. Because the student interface is designed to be accessible via smartphone, students will be able to interact with the system using the devices they already own and use daily without any additional equipment investment.

Officer acceptance is a more nuanced consideration, since departmental staff must be willing to change established working habits and engage consistently with a new system for the workflow to function as intended. The deployment plan described in Chapter Three addresses this through a phased rollout beginning with a pilot involving the Library, followed by structured training for all officers prior to full deployment. The training will cover the core tasks required of each officer role, and printed quick reference guides will be available for ongoing support. The relatively simple and task-focused nature of the officer interface, which presents only the actions relevant to the officer's role, is designed to minimise the learning curve and reduce the risk of resistance. The operational feasibility of the SGCS is therefore assessed as strong.

4.5.4 Schedule Feasibility

Schedule feasibility evaluates whether the project can be completed within the proposed timeline. The SGCS is to be developed between February and August 2026, a period of seven months. As the Gantt chart in Appendix A of the proposal illustrates, the phases of the project are distributed across this period with deliberate overlap where appropriate. Requirement elicitation is complete in April, with analysis and design running through May and June. Development occupies June and July, testing extends through July and August, and deployment and final report writing are scheduled for August.

The scope of the system has been deliberately bounded to ensure deliverability within this timeframe. The SGCS focuses on the six-department sequential clearance workflow,

notifications, transcript handling, reporting, and certificate generation. It does not attempt to replace other institutional systems such as fee payment, examination management, or student registration, each of which would significantly expand the scope and timeline. Given the defined scope, the available development period is sufficient. The technical stack, consisting of familiar standard web technologies, does not introduce a steep learning curve. Schedule feasibility is therefore positive.

4.5.5 Legal and Ethical Feasibility

Legal and ethical feasibility addresses the obligations and standards that the SGCS must comply with in handling student information. The system will collect, store, and process personal data including student names, registration numbers, contact details, financial clearance status, and academic records. In Kenya, the Data Protection Act (2019) establishes the legal framework governing the processing of personal data and imposes obligations on data processors to ensure that personal information is collected for specified, explicit purposes, stored securely, and not shared with unauthorised parties. The SGCS will comply with these requirements through a combination of role-based access control, which ensures that each user can only access information relevant to their function, password-protected accounts with secure session management, and encrypted data transmission using HTTPS.

Ethically, the system is designed to serve students' interests by making the clearance process more transparent, accessible, and fair. The audit trail functionality, which logs every action taken in the system including approvals, rejections, and timestamps, ensures that disputes can be resolved on the basis of factual records rather than conflicting recollections. Students will have clear visibility into the status of their own clearance at all times, and the notification system will ensure they are informed of actions taken on their case without needing to make enquiries. The system does not introduce any mechanism for discriminatory treatment: all students are processed through the same workflow, and the approval criteria are determined by institutional policy rather than by the system itself. The SGCS is therefore both legally compliant and ethically sound.

4.5.6 Feasibility Study Conclusion

Having examined the proposed Students' Graduation Clearance System from technical, economic, operational, schedule, legal, and ethical perspectives, the conclusion of this feasibility study is unequivocal: the project is feasible and its implementation is beneficial. The technology required is widely available, affordable, and well within the capabilities of the

development team. The system addresses a clearly documented institutional need and is designed in a manner that supports adoption by its intended users. The project timeline is realistic given the defined scope, and the system's handling of personal data is consistent with applicable legal and ethical standards. There is no dimension of feasibility that presents a significant barrier to the project proceeding.

4.6 Requirements Analysis

The requirements analysis presented in this section is the product of the document analysis and questionnaire activities described in Chapter Three, supplemented by a thorough examination of the five related systems reviewed in Chapter Two. It establishes, in precise and verifiable terms, what the proposed SGCS must do, the standards it must meet, and the data it must manage. Requirements have been categorised into three groups: functional requirements, which specify the capabilities the system must provide; non-functional requirements, which define the performance and quality standards it must achieve; and data requirements, which identify the information the system must store and manage.

4.6.1 Functional Requirements

FR1 – User Authentication: The system must provide a secure login mechanism for all user types, including students, departmental officers, the Director, the Registrar, and the system administrator. Each login must be verified against stored credentials, and access must be granted only to authenticated users. Sessions must expire after a defined period of inactivity, and users must be able to reset their passwords through a secure email-based process.

FR2 – Student Registration and Profile Management: The system must allow students to create and maintain a profile containing their registration number, full name, programme, institute, email address, phone number, and graduation year. Student profiles must be pre-populated where data can be obtained from the institutional student records, and students must be able to verify and update their contact information.

FR3 – Clearance Initiation: Authenticated students must be able to submit a clearance request through the system. The request must capture the student's details and the target graduation year, and must initiate the sequential departmental workflow beginning with the Institute Director.

FR4 – Sequential Departmental Approval Workflow: The system must enforce a defined approval sequence: Director, Library, Dean of Students, Administration/University Store,

Finance Office, and Academic Registrar. A clearance request must not be presented to a subsequent department until the preceding department has recorded an approval. If a department rejects a request, the student must be notified, and the workflow must pause until the student resolves the outstanding issue and the department re-approves.

FR5 – Departmental Officer Actions: Each departmental officer must be able to log into the system, view the list of clearance requests assigned to their department, review the details of individual requests, record an approval with a timestamp, or record a rejection with a mandatory comment explaining the reason. Officers must also be able to update a rejection to an approval once the underlying issue is resolved.

FR6 – Real-Time Status Tracking: Students must be able to view, from their dashboard at any time, the current status of their clearance request for each department in the workflow. The status for each department must be displayed as one of: Pending, In Review, Approved, or Rejected. The overall clearance status must reflect the furthest department the student has reached in the sequence.

FR7 – Automated Notifications: The system must send automated email notifications to students at each of the following events: clearance request received, departmental approval granted, departmental rejection with reason, and final clearance certificate available for download. Officers must receive notifications when a new clearance request is assigned to their department.

FR8 – Transcript Request and Upload Module: The system must allow students to submit a request for an official academic transcript as part of the clearance process. The Academic Registrar or designated officer must be able to upload the completed transcript file, which the student can then download from within the system.

FR9 – Financial Clearance Verification: The Finance Officer must be able to record whether a student has cleared all financial obligations. If an outstanding balance exists, the officer must be able to enter the amount and record the date on which it was settled, after which the workflow advances.

FR10 – Digital Clearance Certificate Generation: Upon final approval by the Academic Registrar, the system must automatically generate a digital clearance certificate containing the student's name, registration number, programme, graduation year, the names and titles of

approving officers for each department, and the date of final clearance. The certificate must be available for download in PDF format from the student's dashboard.

FR11 – Reporting Dashboard: The Director must have access to a reporting dashboard displaying clearance statistics for students within their institute. The Academic Registrar must have access to institution-wide reports showing the number of students at each stage of the clearance workflow, the number fully cleared, the number with pending or rejected steps, and summary data by department and programme. Reports must be exportable.

FR12 – Audit Trail: The system must maintain an immutable log of all actions taken within the system, including the user, action type, affected record, and timestamp. Audit records must be accessible to the system administrator and to the Academic Registrar for the purpose of resolving disputes or reviewing process compliance.

4.6.2 Non-Functional Requirements

NFR1 – Security: The system must protect all data through HTTPS encryption for all communications. Passwords must be stored as salted hashes using a current cryptographic standard. Role-based access control must ensure that no user can access functions or data outside their defined role. The system must be protected against common web vulnerabilities including SQL injection and cross-site scripting.

NFR2 – Reliability: The system must be capable of sustained, consistent operation without data loss or corruption. Database transactions must be atomic, ensuring that a partial approval cannot be recorded. Backup procedures must be in place to prevent data loss.

NFR3 – Availability: The system must be available during all standard working hours at Tangaza University, with planned maintenance windows scheduled outside these periods and communicated to users in advance. The web hosting environment must provide sufficient uptime guarantees for institutional use.

NFR4 – Usability: All interfaces must be designed to be intuitive for users with basic web browsing skills. Students should be able to initiate a clearance request without any prior training. Officers should be able to process approvals and rejections after a single training session. Error messages must be informative and guide users toward resolution.

NFR5 – Performance: Pages must load within three seconds under normal network conditions. The system must support concurrent use by at least one hundred simultaneous users, which is

sufficient for the scale of Tangaza University's graduating cohort, without performance degradation.

NFR6 – Scalability: The system architecture must be designed to support an increase in the number of concurrent users and records without requiring fundamental redesign. Database schema and query design must be optimised to support growth.

NFR7 – Maintainability: Code must be structured, commented, and documented to a standard that allows another web developer to understand, maintain, and extend the system without requiring involvement of the original developer. Module separation must follow standard PHP MVC practices where applicable.

NFR8 – Responsiveness: The student-facing and officer-facing interfaces must render correctly and remain fully functional on screen widths ranging from 360px on a smartphone to 1920px on a desktop monitor, using CSS responsive design techniques.

4.6.3 Data Requirements

The SGCS must manage several distinct categories of information, each with specific integrity and access requirements. Student data includes registration numbers, full names, institute affiliation, programme, year of study, contact details, and password credentials. This data forms the foundation of the system and must be stored with uniqueness constraints on registration numbers to prevent duplication. Clearance records must capture the student identifier, the date of initiation, the current workflow stage, and the overall clearance status. Each departmental approval record must store the department identifier, the officer who processed it, the action taken, the timestamp, and any comments recorded.

Officer records must include the officer's full name, department, role designation, login credentials, and contact information. Notification logs must preserve a record of each automated message generated by the system, including the recipient, the message type, the content, and the timestamp of dispatch, so that any failure to receive a notification can be investigated. Transcript files uploaded by the Registrar must be stored securely and linked to the relevant student record, with access restricted to the student concerned and authorised staff.

Database integrity must be maintained through the use of foreign key constraints that enforce referential integrity between related tables, unique constraints that prevent duplicate records where applicable, not-null constraints on required fields, and appropriate data types for all

stored values. The database will be normalised to Third Normal Form to eliminate redundant data and ensure that updates are applied consistently across all affected records.

4.7 System Analysis Tools and Models

The SSAD methodology requires the use of standardised graphical tools to represent the current and proposed system processes in a structured and communicable form. This section describes the principal analysis models produced for the SGCS, explains the purpose and content of each, and contextualises their findings within the broader analysis.

4.7.1 Context Diagram

A context diagram, also referred to as a Level 0 DFD, provides the highest-level view of a system by representing it as a single process and showing all external entities that interact with it, together with the data flows passing between those entities and the central process. Its purpose is to establish the boundary of the system clearly: everything inside the boundary is part of the system being designed; everything outside is an external entity that the system communicates with but does not control.

For the current manual clearance system at Tangaza University, the context diagram shows the central process as the Manual Clearance Process, with four external entities interacting with it. The Student submits the clearance form to the system and receives the completed, signed form in return. Each of the six Departmental Officers receives the physical form and returns it with a signature or a rejection note. The Academic Registrar receives the fully signed form and issues the final clearance confirmation. No data stores are shown at this level, as the context diagram is concerned only with the system boundary and the flows crossing it.

For the proposed SGCS, the context diagram replaces the single Manual Clearance Process with the SGCS web application. The external entities remain structurally similar, but the data flows are digital. The Student submits login credentials and a clearance request to the SGCS and receives status updates, notifications, and the clearance certificate. Each Departmental Officer submits approval or rejection decisions and receives clearance request data from the system. The Director and Academic Registrar access reports generated by the system. The System Administrator manages user accounts and system configuration. The context diagram for the proposed SGCS makes visible the significant expansion in the richness and immediacy of information exchange that the digital system enables.

4.7.2 Level 1 DFD

The Level 1 DFD decomposes the single process shown in the context diagram into the major sub-processes that make up the system, together with the data stores and the flows connecting them. For the proposed SGCS, the Level 1 DFD identifies six major processes: Process 1.0 – User Authentication, in which the system validates login credentials against the Users data store; Process 2.0 – Initiate Clearance, in which the student submits a clearance request that is stored in the Clearance Requests data store; Process 3.0 – Departmental Processing, in which departmental officers retrieve, review, and act on pending clearance requests; Process 4.0 – Notification Dispatch, in which the system generates and sends notifications at each stage of the workflow; Process 5.0 – Transcript Management, in which the Registrar uploads and the student downloads academic transcripts; and Process 6.0 – Report Generation, in which the system aggregates clearance data and produces reports for the Director and Registrar.

The data stores at this level include the Students store, the Users store, the Clearance Requests store, the Approval Records store, the Notifications Log, and the Transcripts store. The flows between processes and stores reflect the requirements identified in Section 4.6: a clearance request created in Process 2.0 is retrieved and updated in Process 3.0; each officer action in Process 3.0 triggers Process 4.0; and the aggregated data in the Clearance Requests and Approval Records stores feeds Process 6.0.

4.7.3 Level 2 DFD

The Level 2 DFD expands Process 2.0 (Initiate Clearance) and Process 3.0 (Departmental Processing) to reveal the sub-processes within each. For Process 2.0, the sub-processes are: 2.1 – Validate Student Eligibility, which checks that the student has been confirmed eligible for graduation by the Academic Programme Leader; 2.2 – Create Clearance Request, which writes the initial request record to the Clearance Requests store with a status of Initiated; and 2.3 – Assign to First Department, which sets the active department in the workflow to the Institute Director and creates the first pending approval record in the Approval Records store.

For Process 3.0, the sub-processes are: 3.1 – Retrieve Pending Requests, in which the officer retrieves all requests assigned to their department; 3.2 – Record Decision, in which the officer records an approval or rejection with comments; 3.3 – Update Workflow State, in which an approval triggers the assignment of the clearance request to the next department in the sequence, or a rejection pauses the workflow and flags the request for student attention; and 3.4 – Trigger Notification, which calls the notification dispatch process with the relevant event

type and recipient details. This level of detail confirms the precise logic that the system's backend must implement to enforce the sequential workflow correctly.

4.7.4 System Users and Actors

The Student is the primary end-user of the SGCS. Students interact with the system to create accounts, initiate clearance requests, track departmental approval status, upload documents where required, submit transcript requests, and download their clearance certificate. Students have no access to other students' records or to departmental management functions.

The Departmental Officer represents the administrative staff member within each of the six clearance departments who is responsible for reviewing and acting on clearance requests assigned to their department. Officers can log into a dedicated interface, view a list of pending requests, approve or reject individual requests with comments, and view the history of their own past decisions. Six officer accounts are required, one per department.

The Director is the head of the student's institute and is the first officer in the clearance sequence. In addition to the standard officer functions, the Director has access to a reporting dashboard that shows the clearance progress of all students enrolled in their institute.

The Academic Registrar holds the final approval position in the clearance workflow and has the broadest reporting access of any user, with visibility into clearance statistics across all departments, programmes, and the full graduating cohort. The Registrar is also responsible for uploading academic transcripts for requesting students and for issuing the final clearance confirmation that triggers certificate generation.

The System Administrator manages the technical and administrative configuration of the SGCS, including the creation and deactivation of user accounts, the maintenance of department and programme data, the review of audit logs, and the management of system settings. The administrator does not participate in the clearance workflow itself but ensures the system environment is correctly configured for each graduation cycle.

4.7.5 Inputs and Outputs

The principal inputs to the SGCS are: login credentials (username and password) submitted by all user types for authentication; clearance request data submitted by students including their registration number, programme, and target graduation year; approval and rejection decisions submitted by departmental officers, including the action type and any associated comments;

transcript files uploaded by the Academic Registrar; and reporting parameters submitted by the Director or Registrar to filter and generate reports.

The principal outputs of the SGCS are: automated email notifications dispatched to students and officers at each significant event in the workflow; the clearance status display visible to students from their dashboard, showing the per-department approval state in real time; the digital clearance certificate in PDF format, generated automatically upon final Registrar approval; downloadable academic transcripts made available to students through the system; and management reports summarising clearance progress by student, department, programme, or graduation cycle, available to the Director and Registrar.

4.7.6 System Flowcharts

The student workflow begins when a student navigates to the SGCS login page and authenticates using their registration number and password. Upon successful login, the student is directed to the clearance dashboard, where they can view their clearance status. If no clearance request is active, the student initiates one by confirming their details and submitting the form. The system validates that the student is eligible for graduation and creates the clearance request, assigning it to the Institute Director. The student then monitors the dashboard as the request moves through each department in sequence, receiving email notifications at each stage. If a department rejects the request, the student reviews the rejection comment, addresses the underlying issue, and the relevant officer updates the record to an approval once satisfied. When the Registrar grants the final approval, the system generates the clearance certificate and notifies the student that it is available for download.

The officer workflow begins with login to the SGCS using role-specific credentials. Upon authentication, the officer is directed to the departmental dashboard, which displays a list of clearance requests currently pending their attention. The officer selects an individual request to review the student's details and any notes from preceding departments. The officer then records a decision: if the student has satisfied all requirements for the department, the officer approves the request, which advances it to the next department in the sequence; if an issue is identified, the officer records a rejection with a mandatory descriptive comment. The system notifies the student of the outcome. In the event of a rejection that is subsequently resolved, the officer can return to the request and update the status to approved, at which point the workflow advances.

The approval sequence workflow describes the overall progression of a clearance request through the six-department chain. A request begins at the Institute Director and cannot advance until the Director approves it. It then proceeds to the Library, then to the Dean of Students, then to Administration/University Store, then to the Finance Office, and finally to the Office of the Academic Registrar. Any rejection at any stage halts the workflow for that student until the issue is resolved and re-approved. The overall clearance status of a student reflects the furthest point they have successfully reached in this sequence.

4.8 Chapter Summary

This chapter has conducted a thorough analysis of the graduation clearance environment at Tangaza University, examining the current manual process in detail and assessing it critically from both strengths and weaknesses perspectives. The feasibility study concluded that the proposed SGCS is technically achievable, economically justified, operationally viable, schedulable within the project timeline, and compliant with applicable legal and ethical standards. A comprehensive requirements specification was developed, covering twelve functional requirements, eight non-functional requirements, and a full account of the system's data requirements. The principal SSAD analysis artefacts, including the context diagram, Level 1 and Level 2 DFDs, actor descriptions, input/output catalogues, and system flowcharts, were described in detail, establishing a rigorous analytical foundation for the design work to follow. Chapter Five now builds directly on this foundation, translating the requirements and models documented here into a detailed design specification for the proposed Students' Graduation Clearance System.

CHAPTER FIVE: SYSTEM DESIGN

5.1 Introduction

System design is the phase of the development process in which the abstract understanding produced during analysis is transformed into concrete, implementable specifications. Where analysis answered the question of what the proposed Students' Graduation Clearance System must do and why, design answers the question of how it will do it. This distinction is important: a requirements specification, however thorough, does not by itself tell a developer which tables to create in the database, how the user interface should be laid out, or in what sequence the application should process a clearance request. Design bridges this gap, providing the structural and logical blueprints from which the system will actually be built.

The design work presented in this chapter follows directly from the analysis findings of Chapter Four and is guided throughout by the SSAD methodology introduced in Chapter Three. Every design decision is traceable to a specific requirement: the database schema reflects the data requirements of Section 4.6.3; the process design models elaborate the functional requirements specified in Section 4.6.1; and the interface prototypes are shaped by the non-functional requirements for usability and responsiveness identified in Section 4.6.2. The chapter begins with a description of the proposed system, proceeds through the conceptual architecture, process design, database design, and input/output design, and concludes with a description of the principal reports the system will generate.

5.2 Description of the Proposed System

The proposed Students' Graduation Clearance System is a web-based application designed to replace the manual, paper-based graduation clearance process at Tangaza University with a centralised, automated, and fully digital alternative. It serves five categories of users: students, departmental officers, the Director, the Academic Registrar, and the system administrator, each of whom accesses the system through a role-specific interface authenticated by a unique username and password.

From the student's perspective, the journey through the SGCS begins when they log in and access their personalised dashboard. If no active clearance request exists, the student can initiate one by confirming the accuracy of their academic profile and submitting a clearance request. The system immediately validates the student's eligibility, creates the request record

in the database, and assigns it to the Institute Director as the first officer in the workflow. The student's dashboard updates to reflect that the request is now pending the Director's review.

As the request moves through the six-department sequence, the student's dashboard updates in real time to show the current approval status for each department. When a department approves the request, the student receives an automated email notification and the dashboard reflects the advance to the next department. When a department rejects the request, the student receives an email containing the officer's explanation, and the relevant department's status on the dashboard changes to Rejected with the comment visible. Once the student has resolved the underlying issue and the officer has updated the record, the workflow resumes. If the student has requested a transcript, they can track the status of that request from the same dashboard and download the completed transcript once it has been uploaded by the Registrar.

When the Academic Registrar grants the final approval, the system automatically generates a digital clearance certificate, notifies the student, and makes the certificate available for download from the dashboard. At this point, the student's clearance is complete, and no further action is required.

For departmental officers, the SGCS provides a dedicated dashboard displaying the queue of clearance requests currently awaiting their attention. Each request displays the student's name, registration number, programme, and any notes or comments from preceding departments in the workflow, giving the officer the context they need to make an informed decision. The officer records their decision, and the system handles the subsequent steps: advancing the workflow, updating the student's dashboard, and dispatching the relevant notifications automatically.

The Director and the Academic Registrar each have access to reporting dashboards that present aggregated clearance statistics. The Director's dashboard focuses on students within their institute, while the Registrar's dashboard provides institution-wide data. Both dashboards support filtering by programme, graduation year, and clearance status, and reports can be exported for use in institutional planning.

5.3 Strengths of the Proposed System

The most transformative strength of the SGCS is the complete automation of the clearance workflow. Rather than relying on students to physically transport a form between offices and on officers to be available for face-to-face interaction, the system handles the progression of

requests through the workflow, the generation of notifications, and the updating of status records without any manual coordination. This eliminates the single largest source of delay in the current process, which is the dependence on physical presence and paper-based communication.

Real-time transparency is another significant strength. Under the current system, students have no way to know where their clearance stands without visiting the campus in person. The SGCS gives every student immediate, accurate visibility into the status of their own clearance at all times, accessible from any internet-connected device. This reduces anxiety, eliminates unnecessary campus trips, and empowers students to manage their clearance process proactively rather than reactively.

The centralisation of all clearance data in a single relational database eliminates the data redundancy and integrity risks that characterise the paper-based system. Every action taken in the system is recorded once and is immediately visible to all authorised users. Updates made by one department are instantly reflected across the system without any risk of inconsistency. The audit trail functionality ensures that every decision is permanently recorded and attributable to a specific officer, providing a reliable basis for resolving disputes and supporting institutional accountability.

From an administrative perspective, the reporting capability of the SGCS addresses a gap that the current system cannot fill. The ability to generate real-time reports on clearance progress, broken down by department, programme, or individual student, gives the Director and Registrar the management information they need to identify bottlenecks, respond to delays, and plan the graduation process with confidence. This represents a qualitative improvement in the institution's capacity to manage the clearance process strategically.

The digital generation of clearance certificates eliminates the need for students to make an additional physical visit to collect a printed document, completing the end-to-end digitisation of the clearance process and ensuring that the efficiency gains of the digital workflow are not undermined by a final manual step.

5.4 Weaknesses of the Proposed System

No system is without limitations, and it is important to acknowledge the genuine constraints of the proposed SGCS in order to plan for them appropriately. The most significant limitation is the system's dependence on internet connectivity. A student who lacks reliable internet access,

or an officer working in a location with intermittent connectivity, will be unable to interact with the system during periods of outage. While internet connectivity in Nairobi is generally reliable, it is not universal, and the university should maintain a documented contingency procedure for processing clearance manually in cases of sustained system unavailability.

The initial training requirement for departmental officers represents a short-term operational challenge. Staff who have processed clearance manually for years must adapt to a new workflow, and the quality of the system's operation during the early deployment period will depend significantly on the effectiveness of the training provided. The phased deployment approach described in Chapter Three mitigates this risk by limiting the initial rollout to the Library and a small group of students, allowing the system and its users to stabilise before full deployment.

As a web-based application, the SGCS is subject to the risk of server downtime, which could make the system unavailable during critical periods such as the weeks immediately preceding graduation. This risk is manageable through the selection of a reliable hosting provider with strong uptime guarantees and the maintenance of regular database backups, but it cannot be eliminated entirely. The cybersecurity risks inherent in any web-based system handling personal data, including phishing attempts, unauthorised access, and SQL injection, must also be managed through robust security practices, regular code audits, and staff awareness training. Finally, some resistance to technological change among officers who are comfortable with the existing manual process is likely, and this must be addressed through inclusive training, accessible documentation, and responsive technical support during the transition period.

5.5 Conceptual Architecture of the Proposed System

The SGCS is designed on a three-tier client-server architecture, which separates the system into three logical layers: the presentation layer, the application logic layer, and the data layer. This separation provides a clean structure that makes the system easier to develop, maintain, and scale.

The presentation layer is the browser-based user interface. Users interact with the system through a standard web browser on any device, whether a desktop computer, a laptop, a tablet, or a smartphone. The interface is built with HTML, CSS, and JavaScript and is designed to be fully responsive, adapting its layout to the screen dimensions of the device being used. The presentation layer sends user inputs to the application layer via HTTP requests and displays the responses returned.

The application logic layer consists of PHP scripts running on the web server. This layer receives requests from the browser, processes them according to the business logic of the SGCS, such as determining whether a student is eligible to initiate clearance, enforcing the sequential workflow, and generating notifications, and formulates responses that are returned to the browser. The application layer also communicates with the data layer to retrieve and store information. PHP's native integration with MySQL and its mature web development ecosystem make it an appropriate technology for this layer.

The data layer consists of a MySQL relational database server that stores all persistent data for the system, including student records, user credentials, clearance request states, approval records, notification logs, and certificate files. MySQL's support for transactions, foreign key constraints, and complex queries ensures the data integrity and relational consistency required by the SGCS workflow. The notification subsystem operates as a component of the application layer, triggered by workflow events in the PHP processing logic and dispatching emails to recipients using the SMTP protocol through a configured mail server or third-party email delivery service.

5.6 Process Design Using Appropriate Tools

5.6.1 Context Diagram

The context diagram for the proposed SGCS, designated as Figure 2 in the list of figures, depicts the system as a single process labelled SGCS Web Application and shows the five external entities that interact with it. The Student entity sends login credentials and clearance requests to the system and receives clearance status updates, email notifications, and the digital clearance certificate in return. The Departmental Officer entity sends login credentials and approval or rejection decisions to the system and receives pending clearance request data and notification triggers. The Director entity interacts with the system in both the officer capacity, processing clearance requests from students in their institute, and in the management capacity, receiving institute-level clearance reports. The Academic Registrar entity sends the final approval decision and uploads transcript files, and receives institution-wide report data from the system. The System Administrator entity sends system configuration data and user management commands to the system. No data stores are shown at this level, as they reside within the system boundary.

5.6.2 DFD Level 1

The Level 1 DFD for the proposed SGCS expands the single context diagram process into seven major processes with their associated data stores and flows. Process 1.0 (Authenticate User) receives login credentials from any user entity, queries the Users data store, and returns either an authenticated session or an error. Process 2.0 (Initiate Clearance) receives a clearance request from the Student, validates it against the Students data store, creates a record in the Clearance Requests store, and writes the first pending record to the Approval Records store. Process 3.0 (Process Departmental Approval) reads pending approval records from the Approval Records store, receives decisions from Officer entities, writes updated records back to the Approval Records store, and passes workflow advancement events to Process 4.0. Process 4.0 (Dispatch Notifications) receives event data from Processes 2.0 and 3.0, writes notification records to the Notification Log store, and dispatches email messages to Student and Officer entities. Process 5.0 (Manage Transcripts) receives transcript upload actions from the Registrar, stores files in the Transcripts store, and provides file retrieval to the Student. Process 6.0 (Generate Reports) reads from the Clearance Requests and Approval Records stores and delivers formatted report data to the Director and Registrar. Process 7.0 (Generate Certificate) reads the completed Clearance Request and Approval Records upon Registrar final approval and writes the certificate document to the Certificates store for Student retrieval.

5.6.3 DFD Level 2

The Level 2 DFD further expands Processes 2.0 and 3.0 to expose their internal logic. Within Process 2.0, sub-process 2.1 (Check Student Eligibility) reads the student's record from the Students store and verifies that the Academic Programme Leader's eligibility confirmation flag has been set. Sub-process 2.2 (Create Request Record) writes a new row to the Clearance Requests table with a status of Initiated and the current date as the submission timestamp. Sub-process 2.3 (Assign to First Department) writes the first Approval Record row for the clearance request, designating the Institute Director's department as the active department with a status of Pending.

Within Process 3.0, sub-process 3.1 (Fetch Queue for Officer) reads all Approval Records assigned to the authenticated officer's department with a status of Pending and returns them to the officer dashboard. Sub-process 3.2 (Record Officer Decision) accepts the officer's action input and writes the decision, the officer identifier, and the timestamp to the Approval Record. Sub-process 3.3 (Advance or Halt Workflow) evaluates the recorded decision: if the action is an approval and the department is not the final one in the sequence, it creates a new Pending Approval Record for the next department; if the department is the final one, it updates the

Clearance Request status to Fully Cleared and triggers Process 7.0; if the action is a rejection, it updates the Clearance Request status to Rejected and triggers Process 4.0 with a rejection event. Sub-process 3.4 (Trigger Notification Event) passes the event type and recipient identifiers to Process 4.0.

5.6.4 Use Case Diagram

The use case diagram for the SGCS defines the interactions between each actor type and the system functions available to them. The Student actor is associated with the following use cases: Register Account, Log In, View Clearance Dashboard, Initiate Clearance Request, View Per-Department Status, Receive Email Notification, Request Transcript, Download Transcript, and Download Clearance Certificate. The Departmental Officer actor is associated with: Log In, View Clearance Request Queue, View Request Details, Record Approval, Record Rejection with Comment, and Update Rejection to Approval. The Director actor extends the Departmental Officer actor with the additional use cases View Institute Clearance Report and Export Report Data. The Academic Registrar actor is associated with: Log In, View Final Approval Queue, Record Final Approval, Upload Transcript, View Institution-Wide Reports, and Export Report Data. The System Administrator actor is associated with: Log In, Create User Account, Deactivate User Account, Manage Department Records, Manage Programme Records, and View Audit Trail.

5.6.5 Flowcharts

The system flowcharts translate the logical workflow of the SGCS into a step-by-step procedural representation that maps decision points and action sequences. The Student Clearance Flowchart begins with the student navigating to the login page and entering credentials. The system checks the credentials against the database: if invalid, an error is displayed and the student is prompted to try again; if valid, the student's dashboard is loaded. The system checks whether an active clearance request exists for the student: if not, the student can initiate one; if yes, the current status is displayed. Upon initiation, the system checks eligibility and either confirms success and begins the workflow, or displays an ineligibility message. As each department processes the request, the dashboard updates. If rejected, the rejection comment is displayed. If all six departments approve, the certificate is generated and the download link appears.

The Officer Workflow Flowchart begins with officer login and presents the departmental queue. The officer selects a request to review and examines the student's details and preceding

department comments. The officer then chooses to approve or reject. An approval triggers workflow advancement and a student notification. A rejection requires the officer to enter a mandatory comment before submission, after which the workflow is halted and a rejection notification is sent to the student. The officer can return to a previously rejected request and update it to an approval once the issue is resolved, which resumes the workflow.

5.6.6 Activity Diagram

The activity diagram for the SGCS maps the complete clearance process as a sequence of activities distributed across multiple swim lanes representing the Student, the System, the Departmental Officer (generalised across all six departments), and the Academic Registrar. The process begins in the Student lane with the activity Log In, which transitions to the System lane activity Authenticate User. Upon successful authentication, control returns to the Student lane for the activity Initiate Clearance Request, which passes to the System lane for Check Eligibility and Create Request. The System lane then triggers the Officer lane with Notify Director of Pending Request.

In the Officer lane, the activity Review Request leads to a decision node: Approve or Reject. An Approve decision passes back to the System lane, which advances the workflow to the next department or, if the current department is the last officer before the Registrar, passes to the Academic Registrar lane. A Reject decision passes to the System lane for Update Request Status and Notify Student, and then to the Student lane for Review Rejection and Resolve Issue, after which the workflow returns to the Officer lane. In the Academic Registrar lane, the activities are Review Final Request and Grant Final Approval. The System lane then performs Generate Certificate and Notify Student, after which the Student lane concludes with Download Certificate. The activity diagram makes visible the parallel nature of the notification activities and the iterative nature of the rejection-resolution cycle.

5.6.7 Sequence Diagrams

The sequence diagram for the clearance initiation process involves four participants: the Student (browser), the SGCS Application Server, the Database, and the Email Service. The interaction begins with the Student sending a POST request to the application server containing the clearance initiation form data. The application server sends a query to the Database to verify the student's eligibility. The Database returns the eligibility flag. If eligible, the application server sends an INSERT query to create the Clearance Request record, followed by a second INSERT to create the first Approval Record. The Database confirms both writes. The

application server then calls the Email Service to dispatch a confirmation notification to the student and a pending request notification to the Institute Director. The Email Service confirms dispatch. The application server returns a success response to the Student's browser, which renders the updated dashboard.

The sequence diagram for the departmental approval process involves the same four participants. The Officer's browser sends a POST request containing the approval decision to the application server. The server sends an UPDATE query to the Database to record the decision on the Approval Record. The Database confirms the update. The server then checks the updated workflow state with a SELECT query to determine whether all approvals are complete or whether the next department should be assigned. If the workflow advances, the server sends an INSERT query for the new Approval Record and dispatches notifications via the Email Service. The server returns a success response to the officer's browser, which refreshes the departmental queue.

5.7 Database Design

The database design for the SGCS has been developed through the entity-relationship modelling process described in Section 3.5 and reflects the data requirements identified in Section 4.6.3. The design prioritises referential integrity, minimal redundancy, and efficient query performance for the access patterns that the application will generate most frequently.

5.7.1 Entity Relationship Diagram

The ERD for the SGCS identifies seven principal entities and the relationships between them. The Student entity represents each individual student enrolled in the graduation clearance process and holds personal and academic profile data. The Users entity covers all system users, including students and officers, and stores authentication credentials. The Departments entity represents the six clearance departments, storing the department name, sequence position in the workflow, and the officer assigned to it. The Clearance Requests entity represents an individual student's clearance application for a specific graduation cycle and holds the overall status of the clearance.

The Approval Records entity represents the individual departmental clearance decision for a given clearance request, linking the request to a department and recording the decision, the officer, the timestamp, and any comments. The Notifications entity logs every notification dispatched by the system, linking it to the triggering event, the recipient, and the dispatch timestamp. The Certificates entity stores the generated clearance certificate linked to the

relevant clearance request and records the generation timestamp and file path. The Transcripts entity stores uploaded academic transcript files linked to the student record and the request that prompted the upload.

The principal relationships are as follows: a Student has one or more Clearance Requests (one-to-many); a Clearance Request has exactly six Approval Records (one-to-many, constrained by the number of departments); each Approval Record is associated with one Department (many-to-one); a Clearance Request may have one Certificate (one-to-one); a Student may have one Transcript per graduation cycle (one-to-one); and a Clearance Request may generate multiple Notifications (one-to-many).

5.7.2 Data Dictionary

The data dictionary below describes each table in the SGCS database, listing the field name, data type, constraints, and purpose.

Table	Field	Data Type	Constraints	Description
students	student_id	INT	PK, AUTO_INCREMENT	Unique student record identifier
students	registration_no	VARCHAR (20)	UNIQUE, NOT NULL	Student registration number
students	full_name	VARCHAR (100)	NOT NULL	Full name as on official documents
students	programme	VARCHAR (100)	NOT NULL	Programme of study
students	institute	VARCHAR (100)	NOT NULL	Institute/Faculty affiliation
students	email	VARCHAR (100)	UNIQUE, NOT NULL	Student email address
students	phone	VARCHAR (20)	NULL	Contact phone number

students	graduation_year	YEAR	NOT NULL	Target graduation year
students	eligible_flag	TINYINT(1)	DEFAULT 0	Eligibility confirmed by Programme Leader
users	user_id	INT	PK, AUTO_INCREMENT	Unique user account identifier
users	username	VARCHAR(50)	UNIQUE, NOT NULL	Login username
users	password_hash	VARCHAR(255)	NOT NULL	Bcrypt-hashed password
users	role	ENUM	NOT NULL	student/officer/director/registrar/admin
users	linked_id	INT	NOT NULL	FK to students or departments table
departments	dept_id	INT	PK, AUTO_INCREMENT	Unique department identifier
departments	dept_name	VARCHAR(100)	NOT NULL	Name of clearance department
departments	sequence_no	INT	NOT NULL	Position in workflow (1 to 6)
departments	officer_user_id	INT	FK users.user_id	Assigned officer for the department
clearance_requests	request_id	INT	PK, AUTO_INCREMENT	Unique clearance request identifier

clearance_requests	student_id	INT	FK students.student_id	Student who owns this request
clearance_requests	submitted_at	DATETIME	NOT NULL	Timestamp of request submission
clearance_requests	overall_status	ENUM	NOT NULL	Initiated/In Progress/Rejected/Cleared
clearance_requests	current_dept_id	INT	FK departments.dept_id	Department currently processing
approval_records	approval_id	INT	PK, AUTO_INCREMENT	Unique approval record identifier
approval_records	request_id	INT	FK clearance_requests	Associated clearance request
approval_records	dept_id	INT	FK departments.dept_id	Department processing this record
approval_records	officer_id	INT	FK users.user_id	Officer who recorded the decision
approval_records	status	ENUM	NOT NULL	Pending/Approved/Rejected
approval_records	comments	TEXT	NULL	Officer comments especially on rejection
approval_records	decided_at	DATETIME	NULL	Timestamp of the decision

notifications	notif_id	INT	PK, AUTO_INCREMENT	Unique notification identifier
notifications	recipient_user_id	INT	FK users.user_id	Notification recipient
notifications	notif_type	VARCHAR (50)	NOT NULL	Event type triggering notification
notifications	message	TEXT	NOT NULL	Notification message content
notifications	sent_at	DATETIME	NOT NULL	Dispatch timestamp
certificates	cert_id	INT	PK, AUTO_INCREMENT	Unique certificate identifier
certificates	request_id	INT	FK clearance_requests	Associated clearance request
certificates	generated_at	DATETIME	NOT NULL	Certificate generation timestamp
certificates	file_path	VARCHAR (255)	NOT NULL	Server path to certificate PDF
transcripts	transcript_id	INT	PK, AUTO_INCREMENT	Unique transcript identifier
transcripts	student_id	INT	FK students.student_id	Owning student record
transcripts	uploaded_by	INT	FK users.user_id	Officer who uploaded the file

transcripts	uploaded_at	DATETIME	NOT NULL	Upload timestamp
transcripts	file_path	VARCHAR (255)	NOT NULL	Server path to transcript PDF

5.7.3 Normalisation

Database normalisation is the process of organising a relational database schema to reduce redundancy and improve data integrity by ensuring that each piece of information is stored in exactly one place. The SGCS database has been designed and verified to satisfy the requirements of Third Normal Form (3NF).

First Normal Form (1NF) requires that every column in every table contains only atomic (indivisible) values and that there are no repeating groups. The SGCS database satisfies 1NF throughout: every field stores a single value, and there are no arrays or comma-separated lists embedded in any column. For example, the clearance process for each student across six departments is not stored as six columns in the students table or as a single delimited string, but as six separate rows in the approval_records table, each linked to the student's clearance request by a foreign key.

Second Normal Form (2NF) requires that a table in 1NF must additionally have no partial dependencies, meaning that every non-key column must depend on the entire primary key, not merely part of it. This requirement is relevant only for tables with composite primary keys. All tables in the SGCS database use single-column surrogate primary keys (auto-incrementing integers), which structurally eliminates the possibility of partial dependencies. The SGCS database therefore satisfies 2NF by design.

Third Normal Form (3NF) requires that a table in 2NF must additionally have no transitive dependencies, meaning that no non-key column should depend on another non-key column. Examination of each table confirms compliance: in the clearance_requests table, for example, the current_dept_id field references the departments table through a foreign key rather than storing the department name directly, which would create a transitive dependency on dept_id. Similarly, the programme and institute details of a student are stored once in the students table and referenced by student_id wherever needed, rather than being repeated in the clearance_requests or approval_records tables.

Third Normal Form was selected as the normalisation target for the SGCS database because it eliminates the redundancy and anomalies that would arise at lower normal forms without introducing the performance costs that can accompany higher normal forms such as Boyce-Codd Normal Form. At 3NF, the database schema is structurally clean, update anomalies are eliminated, and query performance remains efficient for the access patterns the application will generate.

5.8 Input and Output Design

The interface design for the SGCS has been guided by the usability and responsiveness requirements specified in Section 4.6.2 and informed by the review of related systems in Chapter Two, particularly the identification of gaps in mobile responsiveness and notification clarity. Each screen has been prototyped with attention to layout clarity, task focus, and minimal cognitive load for users who may not be highly experienced with web applications.

5.8.1 Landing Page

The landing page is the first screen a user encounters when navigating to the SGCS URL. Its purpose is to introduce the system, convey institutional legitimacy, and direct users to the login function. The design features the Tangaza University logo prominently at the top, accompanied by the system title ‘Students’ Graduation Clearance System’ and a brief one-sentence description of the system’s purpose. A centred ‘Login’ button in the university’s primary colour scheme provides the single primary call to action. The page uses a clean, uncluttered layout with institutional photography in the background, chosen to convey the academic environment without distracting from the login prompt. A footer contains contact information for the Registrar’s Office and a link to technical support. The landing page is fully responsive and renders appropriately on mobile devices.

5.8.2 Login Page

The login page presents a centred card layout containing two input fields, Username and Password, a ‘Login’ button, and a ‘Forgot Password?’ link. Input validation provides immediate feedback if either field is left empty before submission. The password field uses masked input as standard. If credentials are invalid, a non-specific error message ‘Invalid username or password. Please try again.’ is displayed to prevent username enumeration. The page does not include a registration link, as student accounts are pre-created by the system administrator using institutional student records. The layout is consistent with the landing page branding and is responsive across device sizes.

5.8.3 Student Dashboard

The student dashboard is the central screen from which a student manages their entire clearance process. It is designed to communicate the most important information at a glance. At the top of the page is a summary card displaying the student's name, registration number, programme, and graduation year, alongside the overall clearance status shown as a coloured badge: green for Fully Cleared, amber for In Progress, and red for Rejected. Below the summary card is the clearance progress tracker, which is a visual horizontal sequence of six steps labelled with the name of each department. Each step displays a status icon: a grey clock for Pending, a blue spinner for In Review, a green tick for Approved, or a red cross for Rejected. Clicking on a Rejected step reveals the officer's comment in an inline panel. A Notifications section on the right side of the screen displays the ten most recent automated messages received by the student. At the bottom of the page, links to the Transcript Request module and the Certificate Download section appear, activated only when the relevant content is available.

5.8.4 Officer Dashboard

The officer dashboard is designed for task efficiency. The primary component is a table listing all clearance requests currently assigned to the officer's department, sorted by submission date with the oldest requests at the top. Each row displays the student's registration number, full name, programme, submission date, and a status badge. An 'Actions' column provides buttons for Approve and Reject. Clicking Reject opens an inline form requiring the officer to enter a mandatory comment before confirming the action, preventing a rejection from being submitted without a reason. Clicking Approve prompts a confirmation dialogue before recording the decision, reducing the risk of accidental approval. A navigation link at the top of the page provides access to the history of past decisions. For the Director, an additional navigation link leads to the institute clearance report.

5.8.5 Clearance Tracking Screen

The clearance tracking screen provides an expanded, detailed view of the student's clearance request. In addition to the visual progress tracker shown on the dashboard, this screen displays a timeline of events in chronological order, showing the date and time of each departmental decision, the name of the approving or rejecting officer, and any comments recorded. This detailed view gives students complete transparency into the history of their clearance and provides the information needed to understand and resolve any rejection. The timeline uses a vertical layout with colour-coded event types: approval events in green, rejection events in red, and system events such as request submission in neutral grey.

5.8.6 Transcript Upload Interface

The transcript management interface is accessible to the Academic Registrar from within the system. It displays a list of students who have submitted transcript requests, with each row showing the student's name, registration number, programme, and the date of the request. An Upload button opens a file selection dialogue restricted to PDF files. Once the file is uploaded and linked to the student record, the row updates to show an Uploaded badge and the upload timestamp, and the student is notified automatically. The student's dashboard is updated to display a Download Transcript link, active immediately upon upload.

5.8.7 Notification Panel

The notification panel is accessible from all authenticated screens via a bell icon in the navigation bar. Clicking the icon opens a dropdown or side panel displaying recent notifications in reverse chronological order. Each notification entry shows the event type (e.g., Department Approved, Rejection Received, Certificate Ready), a brief descriptive message, and the time elapsed since dispatch. Unread notifications are displayed in bold. Clicking a notification navigates the user to the relevant screen. The panel is designed for quick scanning and provides immediate visibility into recent activity without requiring navigation away from the current page.

5.8.8 Certificate Download Page

The certificate download page becomes active on the student's dashboard only after the Academic Registrar has granted final approval and the system has generated the clearance certificate. The page displays the student's name, registration number, programme, graduation year, and the date of final clearance in a formatted summary, along with a prominent Download Certificate button. Clicking the button initiates the download of the PDF clearance certificate. The certificate itself is formatted on a Tangaza University letterhead and contains the student's details, the names and titles of the approving officers for each of the six departments, the date of each approval, and the date of final clearance. A unique certificate reference number is included for verification purposes.

5.8.9 Reports Dashboard

The reports dashboard is accessible to the Director and Academic Registrar. It presents a summary panel at the top showing headline statistics: total students eligible for graduation, number fully cleared, number in progress, and number with at least one rejection. Below the summary, a filterable table allows the user to query clearance records by programme,

department, clearance status, and date range. The table can be sorted by any column and exported to a CSV file for use in external reporting tools. A bar chart visualisation shows the distribution of clearance statuses across departments, making it immediately visible which departments are processing efficiently and which are accumulating a backlog. For the Academic Registrar, an additional tab provides an audit trail view that displays a log of all system actions with filtering by user, action type, and date range.

5.9 Sample Reports

The SGCS reporting module is designed to give the Director and Academic Registrar the management information they need to oversee the clearance process effectively. Four standard reports are generated by the system and are available on demand from the reports dashboard.

The Clearance Progress Report provides a real-time snapshot of the clearance status of the entire graduating cohort. It lists every student who has initiated a clearance request, their programme and institute, the date of initiation, the current stage in the workflow (expressed as the name of the department currently processing their request), and the overall status. The report can be filtered by graduation year, programme, institute, or overall status. It is designed to give the Registrar a complete picture of where the cohort stands at any point during the clearance period and to flag any students who have been in a particular stage for longer than a defined number of days, suggesting a potential delay requiring intervention.

The Students Pending Approval Report lists students whose clearance requests are currently awaiting a decision from one or more departments. For each student, the report shows which department has the pending request and for how long it has been awaiting action. This report is particularly useful for identifying bottlenecks: if a large number of requests are pending at a single department, the report makes this pattern immediately visible, allowing the Registrar to follow up with the relevant officer. The report can be filtered by department to allow each officer or department head to view only the requests pending their action.

The Cleared Students Report lists all students who have obtained final approval from the Academic Registrar and are eligible to have their clearance certificate issued. It displays the student's name, registration number, programme, date of final clearance, and whether the certificate has been downloaded. This report supports the institutional graduation planning process by providing an accurate, up-to-date count of students who are fully cleared and ready for graduation, and by flagging any students who have not yet downloaded their certificate.

The Department Performance Report provides a summary of each department's processing activity over a defined period. For each department, it shows the number of requests reviewed, the number approved, the number rejected, the average time between request assignment and decision, and any requests that have been pending for longer than the acceptable response window. This report supports accountability and operational improvement by making departmental performance data visible to management, without requiring manual data collection from individual offices.

5.10 Chapter Summary

This chapter has presented a comprehensive system design specification for the proposed Students' Graduation Clearance System, translating the requirements and analytical models of Chapter Four into a detailed blueprint ready for implementation. The description of the proposed system established the end-to-end digital clearance journey for students and officers. The conceptual architecture articulated the three-tier client-server model underpinning the system. The process design section produced context diagrams, Level 1 and Level 2 DFDs, a use case diagram, flowcharts, an activity diagram, and sequence diagrams, collectively providing a complete and multi-perspective representation of the system's logic. The database design section presented the ERD, a comprehensive data dictionary for all seven tables, and a normalisation analysis demonstrating compliance with Third Normal Form. The input and output design section described the purpose, layout, and functionality of each principal interface screen, from the landing page through to the reports dashboard, using the language of prototyping to convey screen designs as though they were product artefacts. Four standard reports were specified to support the Director's and Registrar's oversight of the clearance process.

Taken together, the analysis and design work presented in Chapters Four and Five provide a rigorous and complete foundation for the implementation phase of the project. Every design decision is traceable to a documented requirement, and every requirement is traceable to a finding from the analysis of the current system or from the literature and related systems reviewed in Chapter Two. The next chapter will describe the implementation of the SGCS, detailing the development of each system component from the design specifications presented here.

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APPENDICES

Appendix A: Project Schedule

The project will run from February 2026 to August 2026, a total of seven months. The Gantt chart below shows when each phase will be active. Some phases overlap intentionally, reflecting the iterative nature of systems development where design and early implementation can proceed concurrently.

Table A1: Project Schedule - Gantt Chart (February to August 2026)

Activity	Feb	Mar	Apr	May	Jun	Jul	Aug
Topic Selection & Proposal Writing	✓	✓					
Literature Review		✓	✓				
Requirement Elicitation			✓				
System Analysis			✓	✓			
System Design				✓	✓		
System Implementation					✓	✓	
System Testing & UAT						✓	✓
Deployment & Report Writing							✓

Appendix B: Project Resources

The following resources will be required to complete the project:

Human Resources

- Student Researcher (Abigael K. Kirimi) - responsible for all research, design, coding, testing, and report writing activities.
- University Supervisor - to provide academic guidance, review deliverables at each stage, and give final approval on the project report.

- Administrative Staff at Tangaza University - officers from the five clearance departments will participate in User Acceptance Testing during the testing phase.
- Final-Year Students at Tangaza University forty students for questionnaire administration; ten students for User Acceptance Testing.

Equipment

- Laptop Computer - AMD Ryzen 5 processor, 8GB RAM, 256GB SSD, Windows 11 operating system. Already owned by the researcher; no purchase required.
- Internet Access - a stable broadband connection of at least 10 Mbps, required for research, development, and version control throughout the project period.

Software Tools (all free or open-source)

- PHP 8.x - server-side scripting language for backend development (open-source, free).
- MySQL - relational database management system (open-source, free).
- XAMPP - local development environment bundling Apache, PHP, and MySQL (open-source, free).
- Visual Studio Code - code editor with PHP and web development extensions (free).
- Git and GitHub - version control and source code repository (free).

Appendix C: Project Budget

The estimated budget for this project is presented in Table C1 below. All figures are in Kenyan Shillings (KES). Because all development tools are open-source and freely available, the primary costs are internet connectivity, web hosting, and documentation. A ten percent contingency allowance has been included to cover unforeseen expenses.

Table C1: Project Budget

S/N	Item	Quantity	Unit Cost	Total
1	Laptop Computer (AMD Ryzen 5, 8GB RAM, 256GB SSD)	1	Already owned	KES 0
2	Internet subscription (6 months)	6 months	KES 1,500/month	KES 9,000

S/N	Item	Quantity	Unit Cost	Total
3	Web hosting - shared hosting plan (6 months)	6 months	KES 800/month	KES 4,800
4	Domain name registration (1 year)	1	KES 1,200	KES 1,200
5	Printing & binding - proposal + final report	2 copies	KES 1,500	KES 3,000
6	Stationery (notebooks, pens)	1 set	KES 500	KES 500
7	Transport to Tangaza University (interviews + UAT)	4 trips	KES 400/trip	KES 1,600
8	PHP / MySQL / XAMPP / VS Code / Git	Open-source	Free	KES 0
9	Contingency (10%)	–	–	KES 2,010
	TOTAL			KES 22,110

The total estimated budget of KES 22,110 is manageable within the constraints of a student-funded project. The largest single cost is internet subscription, which is essential for research, development, and version control throughout the project period. All development tools are available free of charge, significantly reducing the overall cost of the project.

Appendix D: Document Analysis - Reference Materials

D1: Sample Questionnaire for Final-Year Students

Dear Respondent,

This questionnaire is part of a final-year Computer Science project at the Catholic University of Eastern Africa, which involves developing a Students' Graduation Clearance System for Tangaza University. Completion of this questionnaire should take no more than ten minutes. All information provided will be treated with strict confidentiality and used solely for academic purposes.

SECTION A: BACKGROUND INFORMATION

1. What is your current year of study? ☐ Final Year ☐ Other: _____
2. What is your gender? ☐ Male ☐ Female ☐ Prefer not to say
3. How do you mainly access online services? ☐ Smartphone ☐ Laptop/Desktop ☐ Both

SECTION B: CURRENT CLEARANCE PROCESS

4. Have you gone through the graduation clearance process at Tangaza University? ☐ Yes ☐ No
5. If yes, how long did the entire process take? ☐ Less than 1 day ☐ 1–2 days ☐ 3–5 days ☐ More than 5 days
6. How would you rate the current manual clearance process? ☐ Very Poor ☐ Poor ☐ Fair ☐ Good ☐ Excellent
7. What challenges did you face? (Tick all that apply) ☐ Long queues ☐ Unavailable officers ☐ Multiple return visits ☐ Lost/misplaced forms ☐ Poor communication about status ☐ Other: _____

SECTION C: PROPOSED ONLINE SYSTEM

8. Would you prefer an online clearance system? ☐ Yes ☐ No ☐ Neutral
9. How important are notifications about clearance progress to you? (1 = Not important, 5 = Very important) ☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5
10. Which feature is most important to you? (Rank 1–5) ☐ Online initiation ☐ Real-time status per department ☐ Email notifications ☐ Digital certificate download ☐ Automated reminders
11. Do you have reliable internet access? ☐ Yes ☐ Sometimes ☐ No
12. Any other comments or suggestions for the proposed system:
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Thank you for your participation.

D2: Sample Questionnaire for Final-Year Students

